Journal of Mathematical Economics 120 (2025) 103155

Contents lists available at ScienceDirect

Journal of Mathematical Economics

journal homepage: www.elsevier.com/locate/jmateco

Unknottedness of graphs of pairwise stable networks & network dynamics

Julien Fixary

Université Mohammed VI Polytechnique, Moroccan Center for Game Theory, Morocco

ARTICLE INFO

Manuscript handled by Editor L.P. Merlino

Keywords:
Pairwise stability
Unknottedness theorem
Network dynamics
Genericity

ABSTRACT

We extend Bich–Fixary's topological structure theorem about graphs of pairwise stable networks. Specifically, we show that certain graphs of pairwise stable networks are not only homeomorphic to their underlying space of societies but are, in fact, ambient isotopic to a trivial copy of this space. This result aligns with Demichelis–Germano's unknottedness theorem and Predtetchinski's unknottedness theorem. Furthermore, we introduce the notion of network dynamics which refers to families of vector fields on the set of weighted networks whose zeros correspond to pairwise stable networks. We leverage our version of the unknottedness theorem to demonstrate that most network dynamics can be continuously connected to one another without introducing additional zeros. Finally, we show that this result has a significant consequence on the indices of these network dynamics at any pairwise stable network — a concept that we connect to genericity using Bich–Fixary's oddness theorem.

1. Introduction

For several years, network formation theory has been gaining importance in economic theory. Informally, a network consists of a set of agents and a set of links, where a link between two agents represents a certain relationship between them (e.g., a network defined by a group of people and their friendship ties, or a network defined by a group of researchers and their co-authorship relationships). Jackson and Wolinsky (1996) introduced the concept of pairwise stability which is widely used in network formation theory in order to predict which unweighted networks are likely to emerge in a strategic setting. In this framework, each agent is assigned a payoff function that represents their linking preferences. Briefly, a network is said to be pairwise stable if no two agents can jointly benefit from forming a new link and no single agent can benefit from severing an existing one. Bich and Morhaim (2020) extended the concept of pairwise stability to weighted networks, i.e., networks in which the strength of each link is represented by a real number between 0 and 1. They also proved the existence of a pairwise stable weighted network for broad classes of payoff functions. Recently, Bich and Fixary (2022) contributed to the study of the topological structure of graphs of pairwise stable networks, which correspond to spaces of the form

$$\{(v, g) : g \text{ is pairwise stable with respect to } v\},$$

where v is a profile of payoff functions for the agents involved in the network formation process — referred to as a "society" — and belongs to a fixed space, while g is a weighted network. One consequence of this contribution (established in the same paper) is that, for large classes of polynomial payoff functions, there generically exists an odd number of pairwise stable networks. These results align with many findings in game theory (see, e.g., Blume and Zame, 1994; Govindan and McLennan, 2001; Govindan and Wilson, 2001; Harsanyi, 1973; Herings and Peeters, 2001; Kohlberg and Mertens, 1986; Mas-Colell, 2010; Wilson, 1971).[1] The central objective of this paper is to further emphasize the parallels between game theory (with the concept of Nash equilibrium) and network formation theory (with the concept of pairwise stability).[2] To this end, we focus on two specific topics, which are briefly outlined below.

Kohlberg and Mertens (1986) provided their famous topological structure theorem which states the following: given a fixed number of players and fixed finite strategy sets, the space

$$\{(u, \sigma) : \sigma \text{ is a mixed-strategy Nash equilibrium of } u\}$$

E-mail address: julien.fixary@um6p.ma.

[1] In this line of research, Bich and Fixary (2024) also established a similar result for large classes of polynomial payoff functions and semialgebraic strategy sets.

[2] However, it is important to note that, by nature, the concept of pairwise stability is not directly comparable to that of Nash equilibrium, as a pairwise stable network is defined over unordered pairs of agents and is characterized by both a cooperative and a non-cooperative aspect. To use the words of Jackson and Wolinsky (1996): "*The formation of a link requires the consent of both parties involved, but severance can be done unilaterally*".

[3] We can also mention the notable work of Demichelis and Germano (2000b) in general equilibrium theory, where they provided significant results concerning the unknottedness of the graph of Walras equilibria and its implications for price dynamics.

https://doi.org/10.1016/j.jmateco.2025.103155
Received 25 November 2024; Received in revised form 14 June 2025; Accepted 25 June 2025
Available online 14 July 2025
0304-4068/© 2025 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

— called the graph of mixed-strategy Nash equilibria associated with the space of finite games — is homeomorphic to the space of finite games itself. Intuitively, this result means that these objects can be continuously deformed into one another (it is worth noting that, topologically speaking, the space of finite games is a much simpler object since it can be identified with some Euclidean space). A few years later, Demichelis and Germano (2002) strengthened Kohlberg–Mertens' theorem by proving an unknottedness theorem. Specifically, they showed that the graph of mixed-strategy Nash equilibria associated with the space of finite games is not only homeomorphic to the underlying space of games, but also ambient isotopic to a trivial copy of this space within a suitable ambient space. The key idea behind this result is that this ambient isotopy consists of a continuous deformation of the entire ambient space, which in turn induces a continuous deformation of the two objects under consideration. This result has important consequences for Nash fields — also called "Nash dynamics" — which are discussed in the same paper (we will return to this point later).[3] An important aspect of this construction is that it is primarily based on the homeomorphism established in Kohlberg–Mertens' theorem (essentially, the idea is to extend their homeomorphism to the ambient space). More recently, Predtetchinski (2009) made significant contributions to the study of topological properties of graphs of Nash equilibria. First, he generalized Kohlberg–Mertens' topological structure theorem to own-strategy C^1 concave games. More precisely, given a fixed number of players and a fixed compact convex strategy set for each player, Predtetchinski proved that the graph of (pure-strategy) Nash equilibria associated with the space of own-strategy C^1 concave games — defined analogously as before — is homeomorphic to the underlying space of games. Second, in the spirit of Demichelis and Germano, Predtetchinski established his own unknottedness theorem. It is worth noting that Predtetchinski's construction of an ambient isotopy also relies heavily on the homeomorphism established in his own topological structure theorem. Now, what about network formation theory? As briefly mentioned at the beginning of this introduction, Bich and Fixary (2022) contributed to the study of the topological structure of graphs of pairwise stable networks. These objects are analogous to graphs of Nash equilibria described above. More precisely, Bich–Fixary's topological structure theorem states that for any "well-behaved" space $\mathcal{V}$ of societies (referred to as "regular"), the graph of pairwise stable networks associated with $\mathcal{V}$ is homeomorphic to $\mathcal{V}$. This result is particularly interesting for two reasons: (i) it strengthens the connection between network formation theory and game theory (it is intriguing that two different research fields, with distinct equilibrium concepts, exhibit similar topological properties); (ii) by analogy with Predtetchinski's result, the considered class of payoff functions is broader (the authors establish a homeomorphism for each regular space of societies rather than for a single specific case). Given the results discussed above, a natural question arises: can an unknottedness result be established in the context of network formation theory? The first goal of this paper is to address this question. Specifically, we strengthen Bich–Fixary's topological structure theorem by showing that the graph of pairwise stable networks associated with any regular space $\mathcal{V}$ is not only homeomorphic to $\mathcal{V}$, but also ambient isotopic to a trivial copy of it within a suitable ambient space — a result we refer to as our *unknottedness theorem*. The general approach follows the frameworks established by Demichelis and Germano, as well as Predtetchinski: it relies on a topological structure theorem (in this case, Bich–Fixary's theorem). However, two main differences arise in our setting. First, because we consider a larger class of payoff functions — still by analogy with Predtetchinski's theorem — it is essential to ensure that the ambient isotopy we construct respects the structure of the underlying space. Specifically, for any regular space $\mathcal{V}$ of societies, the ambient isotopy must map elements of $\mathcal{V}$ into $\mathcal{V}$. Simply following Predtetchinski's approach does not guarantee this property a priori. Second, unlike previous unknottedness results, which concern games and Nash equilibria, our framework involves societies and pairwise stability. This necessitates a distinct approach to constructing the ambient isotopy. Specifically, whereas in game theory the ambient isotopies in previous results operate on strategy profiles — mapping each profile to another — our setting involves networks, and the ambient isotopy must thus map each network to another network. Therefore, while our method bears some resemblance to Predtetchinski's, specific adjustments are required to ensure that the resulting ambient isotopy is well-defined in our context.

As mentioned above, still in their 2002 paper, Demichelis and Germano used their unknottedness result to establish some fundamental properties of Nash dynamics. This notion has been studied extensively in the case of mixed extensions of finite strategic-form games (see, e.g., Demichelis and Germano, 2000a, 2002; Govindan and Wilson, 1997; Gul et al., 1993; Ritzberger, 1994).[4] The second objective of this paper is twofold. First, we introduce the concept of network dynamics, a notion inspired by Nash dynamics. More precisely, a network dynamics is defined as a family of vector fields on the set of networks whose zeros coincide with the set of pairwise stable networks associated with societies in any regular space. This notion is, in a sense, more general than that of Demichelis and Germano, as we do not merely work with unweighted networks and some underlying "mixed extension" but rather with weighted networks and broader spaces of societies. Second, as Demichelis and Germano did in their framework, we leverage our unknottedness theorem to prove the second main result of this paper, called *dynamics equivalence theorem*, which states that any two well-behaved network dynamics are homotopic within the set of all network dynamics on any regular space of societies. In other words, these network dynamics can be continuously connected without introducing additional zeros. An important corollary follows from this theorem, the *indices equality theorem*, which asserts that the index at any pairwise stable network does not depend on the choice of the (still well-behaved) network dynamics. Finally, we establish a connection with genericity through our *semialgebraic indices equality theorem*. These results naturally raise a first question: what are the similarities and differences between our findings and those of Demichelis and Germano? Unsurprisingly, the answer mirrors the earlier discussion on the similarities and differences between Predtetchinski's unknottedness theorem and our own. As mentioned above, the main similarity lies in the overall structure of the approach: as in the work of Demichelis and Germano, the properties of our network dynamics (e.g., the dynamics equivalence theorem) are primarily derived from our unknottedness result. However, key differences include our consideration of a broader class of payoff functions and the fact that we work with societies rather than games. This requires careful verification that the constructions involved — particularly in the dynamics equivalence theorem — remain well-behaved in this broader context. The approach adopted in this paper to formalize and prove our results relies primarily on vector bundles and differential geometry. Another question that naturally arises is: what does the notion of network dynamics bring to the field of network formation theory? To address this second question, it is useful to consider two complementary perspectives. First, from a theoretical viewpoint: over the past years, evolutionary processes in *unweighted* networks have been extensively studied (see, e.g., Jackson and Watts, 2001, 2002; Watts, 2001). The objective of this paper is to provide a theoretical foundation for evolutionary processes in *weighted* networks, adopting a more differential-geometric perspective. Classical dynamical network formation models (such as those mentioned above) are typically based on Markovian dynamics or on processes involving the sequential addition or deletion of links over time. The notion of network dynamics offers an alternative, providing a continuous and smoother description of network evolution. Furthermore, this approach facilitates the analysis of rest points, which in this framework correspond to pairwise stable networks. Unlike models based on local

[4] For a reference on Nash dynamics, see Laraki et al. (2019), pp. 84–85, or Demichelis and Germano (2002).

rules and probabilistic link formation/destruction, network dynamics can reveal a more global structure underlying the dynamical formation process. Second, regarding the conceptual motivation of this paper: a crucial aspect of our philosophy is the connection between network formation theory and game theory — at least from a theoretical standpoint. Thus, the primary goal here is not merely to provide direct applications of the notion of network dynamics but rather to offer compelling arguments supporting this theoretical bridge. One might argue that this paper is somewhat "disconnected" from network formation theory from a purely practical perspective. However, this viewpoint changes when considering the historical development of Nash equilibrium. In his thesis, Nash introduced his concept from a purely theoretical standpoint. While he undoubtedly recognized its potential applications — hence his mention of economics and bargaining theory — practical implementations only emerged decades later, particularly through the works of Aumann and Harsanyi. Similarly, the notion of network dynamics could prove useful in various applied contexts. For instance, it could help describe how network structures between firms evolve based on their strategic preferences. In many industries, firms establish and adjust relationships (such as alliances, supplier agreements, or R&D partnerships) according to expected benefits and competitive pressures. A dynamic, differential-geometric approach could provide insights into the long-term stability and evolution of these inter-firm connections. Another potential application lies in understanding how the degree of connectivity between individuals affects the diffusion of a technology, particularly in online networks. Traditional models often rely on discrete-time adoption processes or stochastic diffusion mechanisms. A continuous approach based on network dynamics could offer a complementary perspective, capturing how small variations in influence or connectivity can gradually shape large-scale adoption patterns. Finally, network dynamics could shed light on how inter-species cooperation evolves under external pressures, such as environmental changes or shifts in resource availability. In ecological and evolutionary studies, the strength of interactions between species (e.g., symbiotic relationships, predation, or competition) is often modeled using fixed or probabilistic frameworks. By considering these interactions within a weighted, continuous dynamics framework, one could explore how cooperative structures emerge, stabilize, or dissolve over time in response to external perturbations. Returning to the present paper, we hope that these examples will serve as a valuable starting point for researchers in applied network formation theory, providing both a theoretical foundation and new avenues for empirical investigation.

This paper is organized as follows. In Sections 3 and 4, we review some elementary concepts from network formation theory, along with the topological structure result of Bich and Fixary (2022), which is a crucial element for the rest of the paper. In Section 5, we present our unknottedness theorem, which is the first main result of the paper. In Section 6, we introduce the notion of network dynamics and provide an explicit example. We also present our dynamics equivalence theorem, which constitutes the second main result of this paper, as well as several previously mentioned corollaries concerning the indices of network dynamics and genericity (our indices equality theorem and its semialgebraic version). Finally, in the appendix, we offer mathematical reminders on general topology, differential calculus, differential geometry, and (real) semialgebraic geometry (Appendix A), along with the proofs of all results of the paper (Appendix B to Appendix F).

2. Notations

Throughout this paper, unless otherwise specified: (*i*) any subset of a topological space is endowed with the induced topology; (*ii*) any finite Cartesian product of topological spaces is endowed with the product topology; (*iii*) any Euclidean space is endowed with the Euclidean topology.

In addition, we provide in Table 1 a list of the key elementary mathematical notations used throughout the core of the paper.

Table 1
Key elementary mathematical notations.

$\lvert A\rvert$	*Cardinality of the set A*
$B\backslash A$	*Relative complement of the set A in the set B*
$\{A \to B\}$, $\mathcal{F}(A,B)$, B^A	*Set of all functions from the set A to the set B*
$\mathrm{im}(f)$	*Image of the function f*
$\mathrm{Gr}(f)$	*Graph of the function f*
id_A	*Identity function on the set A*
$\mathcal{C}^0(X,Y)$	*Set of all continuous functions from the topological space X to the topological space Y*
$\mathrm{int}(A)$	*Interior of the set A (included in some topological space)*
∂A	*Boundary of the set A (included in some topological space)*
$\mathrm{T}M$	*Tangent bundle of the smooth manifold M*
$\mathfrak{X}(M)$	*Vector space of all vector fields on the smooth manifold M*
$\mathrm{Span}(A)$	*Linear span of the set A (included in some vector space)*
$\dim(S)$	*Dimension of the semialgebraic set S*

3. Graphs of pairwise stable networks

First, let us recall some basic definitions and notations from network formation theory.

Definition 3.1. A *set of agents* is a finite set I such that $|I| > 1$. For every set I of agents, the set $\mathbf{L} = \{\{i,j\} : (i,j) \in I \times I, i \neq j\}$ is called the *set of links* and the set $\mathbf{G} = \{\mathbf{L} \to [0,1]\}$ is called the *set of (weighted) networks*.

Definition 3.2. A *(weighted) society* is a couple $(I,(v_i)_{i\in I})$ consisting of a set I of agents and a family $(v_i)_{i\in I}$ of functions from $\mathbf{G}$ to $\mathbb{R}$, where for every $i \in I$, v_i is called the *payoff function of agent i*.

Throughout this paper, we consider a fixed set I of agents.

Remark 3.1. For every set I of agents, the set of all societies whose set of agents is equal to I can be identified with the set

$$\mathcal{S} = \{I \to \{\mathbf{G} \to \mathbb{R}\}\}.$$

In other words, since the set of agents is fixed in our case, we will refer to the term "society" as the underlying profile of payoff functions.

Notations.

- The cardinality of $\mathbf{L}$ is denoted by ℓ.
- Every link $\{i,j\} \in \mathbf{L}$ is denoted by ij.
- For every network $g \in \mathbf{G}$ and every link $ij \in \mathbf{L}$, $g(ij)$ is denoted by g_{ij}.
- For every link $ij \in \mathbf{L}$, $\mathbf{L}_{-ij} = \mathbf{L}\backslash ij$ and $\mathbf{G}_{-ij} = \{\mathbf{L}_{-ij} \to [0,1]\}$.
- For every link $ij \in \mathbf{L}$, every $g_{-ij} = (g_{kl})_{kl\neq ij} \in \mathbf{G}_{-ij}$ and every $w \in [0,1]$, we define the network $\gamma = (w, g_{-ij}) \in \mathbf{G}$ by setting $\gamma_{kl} = g_{kl}$ for every $kl \neq ij$, and $\gamma_{ij} = w$.
- For every network $g \in \mathbf{G}$ and every link $ij \in \mathbf{L}$, we denote $g_{-ij} = (g_{kl})_{kl\neq ij} \in \mathbf{G}_{-ij}$.

Pairwise stability is one of the most important concepts in network formation theory. It was introduced by Jackson and Wolinsky (1996) for unweighted societies,[5] and was later extended by Bich and Morhaim (2020) to weighted societies.

Definition 3.3. Let $v \in \mathcal{S}$ be a society. A network $g \in \mathbf{G}$ is *pairwise stable with respect to v* if, for every $ij \in \mathbf{L}$, the two following conditions hold:

1. For every $w \in [0, g_{ij})$, $v_i(w, g_{-ij}) \leq v_i(g)$ and $v_j(w, g_{-ij}) \leq v_j(g)$.

[5] For every set I of agents, the *set of unweighted networks* is defined by $\mathbf{G}_u = \{\mathbf{L} \to \{0,1\}\}$, and an *unweighted society* is a couple $(I,(v_i)_{i\in I})$ consisting of a set I of agents and a family $(v_i)_{i\in I}$ of functions from $\mathbf{G}_u$ to $\mathbb{R}$.

2. For every $w \in (g_{ij}, 1]$, $v_i(w, g_{-ij}) \leq v_i(g)$ or $v_j(w, g_{-ij}) \leq v_j(g)$.

Definition 3.4. Let $\mathcal{V} \subset \mathcal{S}$ be a set of societies. The *pairwise stable networks correspondence associated with* $\mathcal{V}$ is the correspondence

$$\begin{cases} \mathcal{V} & \twoheadrightarrow & \mathbf{G} \\ v & \mapsto & \{g \in \mathbf{G} : g \text{ is pairwise stable with respect to } v\} \end{cases}.$$

The graph of the pairwise stable networks correspondence associated with $\mathcal{V}$ is called the *graph of pairwise stable networks associated with* $\mathcal{V}$ and is denoted $\mathbf{P}_{\mathcal{V}}$, i.e.,

$$\mathbf{P}_{\mathcal{V}} = \{(v, g) \in \mathcal{V} \times \mathbf{G} : g \text{ is pairwise stable with respect to } v\}.$$

The projection from $\mathbf{P}_{\mathcal{V}}$ to $\mathcal{V}$ is denoted by $\pi_{\mathcal{V}}$, and the projection from $\mathbf{P}_{\mathcal{V}}$ to $\mathbf{G}$ is denoted by $\pi_{\mathbf{G}}$.

4. Reminders on the topological structure of graphs of pairwise stable networks

The payoff functions considered in this paper must satisfy certain differentiability and concavity properties.

Definition 4.1. For every $i \in I$,

$$\mathcal{C}_i = \{v_i \in \mathcal{F}(\mathbf{G}, \mathbb{R}) : \forall j \neq i, \forall g_{-ij} \in \mathbf{G}_{-ij}, v_i(\cdot, g_{-ij}) \text{ is concave}\},$$

$$\mathcal{D}_i = \{v_i \in \mathcal{C}^0(\mathbf{G}, \mathbb{R}) : \forall j \neq i, \partial_{ij} v_i \text{ exists and is continuous}\},$$

where for every $v_i \in \mathcal{C}^0(\mathbf{G}, \mathbb{R})$ and every $j \neq i$,[6]

$$\partial_{ij} v_i : g = (g_{ij}, g_{-ij}) \in \mathbf{G} \mapsto \frac{\partial v_i}{\partial g_{ij}}(g) \in \mathbb{R}.$$

Additionally, we define

$$\mathcal{C} = \prod_{i \in I} \mathcal{C}_i \text{ and } \mathcal{D} = \prod_{i \in I} \mathcal{D}_i.$$

Definition 4.2. For every $i \in I$, the vector space $\mathcal{D}_i$ (with its usual operations) is endowed with the following norm:

$$\|\cdot\|_i : v_i \in \mathcal{D}_i \mapsto \max\{\max\{\|v_i\|_\infty, \|\partial_{ij} v_i\|_\infty\} : j \neq i\} \in \mathbb{R},$$

where for every $v_i \in \mathcal{D}_i$ and every $j \neq i$,[7]

$$\|v_i\|_\infty = \sup_{g \in \mathbf{G}} |v_i(g)| \text{ and } \|\partial_{ij} v_i\|_\infty = \sup_{g \in \mathbf{G}} |\partial_{ij} v_i(g)|.$$

Furthermore, any subset of $\mathcal{D}_i$ is endowed with the induced topology.

For every $i \in I$, we consider the set of payoff functions of agent i that are affine in $(g_{ij})_{j \neq i}$ and depend only on these weights.

Definition 4.3. For every $i \in I$,

$$\mathcal{A}_i = \{g \in \mathbf{G} \mapsto \sum_{j \neq i} b_{ij} g_{ij} + c_i \in \mathbb{R} : \forall j \neq i, b_{ij} \in \mathbb{R}, c_i \in \mathbb{R}\}.$$

Additionally, we define

$$\mathcal{A} = \prod_{i \in I} \mathcal{A}_i.$$

We now recall the notion of regular set of societies, introduced in Bich and Fixary (2022), which is central to Bich–Fixary's topological structure theorem (see Theorem 4.1 below).

Definition 4.4. A set $\mathcal{V} \subset \mathcal{S}$ of societies is called *regular* if it satisfies the following three conditions:

1. **Concavity:** $\mathcal{V} \subset \mathcal{C}$.
2. **Differentiability:** $\mathcal{V} \subset \mathcal{D}$.
3. **$\mathcal{A}$-invariance:** $\mathcal{V} + \mathcal{A} = \mathcal{V}$.

The following result describes the topological structure of the graph of pairwise stable networks associated with any regular set of societies.

Theorem 4.1 (Bich and Fixary, 2022). *For every regular set $\mathcal{V}$ of societies, the projection $\pi_{\mathcal{V}} : \mathbf{P}_{\mathcal{V}} \to \mathcal{V}$ is properly homotopic to some homeomorphism $\eta_{\mathcal{V}} : \mathbf{P}_{\mathcal{V}} \to \mathcal{V}$ with inverse $\rho_{\mathcal{V}} : \mathcal{V} \to \mathbf{P}_{\mathcal{V}}$.*

Consider a regular set $\mathcal{V} \subset \mathcal{S}$ of societies. The $\mathcal{A}$-invariance assumption on $\mathcal{V}$ might initially seem somewhat unusual. However, it can be interpreted as a form of "stability under affine addition" of the payoff functions. In simple terms, this condition ensures that the homotopy in Theorem 4.1 between the projection $\pi_{\mathcal{V}}$ and the homeomorphism $\eta_{\mathcal{V}}$ is well-defined. This is particularly relevant to the construction of Predtetchinski's homotopy in his topological structure theorem (Predtetchinski, 2009). Nevertheless, as noted in Bich and Fixary's work (Bich and Fixary, 2022), this assumption is not particularly restrictive and allows for the construction of meaningful examples in the context of network formation theory.

As mentioned in the introduction of this paper, Kohlberg and Mertens (1986) provided a similar result in 1986 for the case of mixed-strategy Nash equilibria in finite strategic-form games. Predtetchinski (2009) generalized Kohlberg–Mertens's structure theorem to Nash equilibria in own-strategy $\mathcal{C}^1$ concave games. More recently, Bich and Fixary (2024) extended Predtetchinski's result to regular sets of games. Additionally, in the same paper, the authors introduced a new generic oddness result for graphs of Nash equilibria in the semialgebraic case.

5. Unknottedness of graphs of pairwise stable networks

In this section, for every set $\mathcal{V} \subset \mathcal{S}$ of societies, we consider an "enlarged" space of $\mathcal{V} \times \mathbf{G}$, which we call the ambient space associated with $\mathcal{V}$.

Notations. For every set $\mathcal{V} \subset \mathcal{S}$ of societies, define $\mathbf{E}_{\mathcal{V}} = \mathcal{V} \times \mathbf{G}$. For every set $\mathcal{V} \subset \mathcal{S}$ of societies and every $g^0 \in \mathbf{G}$, define $\mathcal{V}^0 = \mathcal{V} \times \{g^0\}$.

Definition 5.1. Let $\varepsilon \in (0, +\infty)$ and $\mathcal{V} \subset \mathcal{S}$ be a set of societies. The *(ε-)ambient space associated with* $\mathcal{V}$ is the topological space

$$\mathbf{E}^{\varepsilon}_{\mathcal{V}} = \mathcal{V} \times \mathbf{G}^{\varepsilon},$$

where

$$\mathbf{G}^{\varepsilon} = \{\mathbf{L} \to [-\varepsilon, 1 + \varepsilon]\}.$$

If there is no ambiguity, $\mathbf{E}^{\varepsilon}_{\mathcal{V}}$ (resp. $\mathbf{G}^{\varepsilon}$) is also denoted $\widetilde{\mathbf{E}}_{\mathcal{V}}$ (resp. $\widetilde{\mathbf{G}}$).

Throughout the rest of this paper, we will consider a fixed real number $\varepsilon > 0$ and a fixed network $g^0 \in \mathbf{G}$.

Definition 5.2. Let $\mathcal{V} \subset \mathcal{S}$ be a regular set of societies. The *unknot associated with* $\mathcal{V}$ *(and* g^0*)* is the topological embedding

$$\begin{cases} \kappa^0_{\mathcal{V}} : & \mathcal{V} & \to & \widetilde{\mathbf{E}}_{\mathcal{V}} \\ & v & \mapsto & (v, g^0) \end{cases}.$$

The *knot of pairwise stable networks associated with* $\mathcal{V}$ *(and* $\rho_{\mathcal{V}}$*)* is the topological embedding

$$\begin{cases} \kappa_{\mathcal{V}} : & \mathcal{V} & \to & \widetilde{\mathbf{E}}_{\mathcal{V}} \\ & v & \mapsto & \rho_{\mathcal{V}}(v) \end{cases}.$$

[6] Since $\mathbf{G}$ is a closed convex subset of $\mathbb{R}^{\ell}$ such that $\mathrm{int}(\mathbf{G}) \neq \emptyset$, for every $i \in I$, every $v_i \in \mathcal{C}^0(\mathbf{G}, \mathbb{R})$ and every $j \neq i$, the function $\partial_{ij} v_i$ is well-defined. Furthermore, for every $i \in I$, every $v_i \in \mathcal{D}_i$, every $j \neq i$ and every $g_{-ij} \in \mathbf{G}_{-ij}$, the function $v_i(\cdot, g_{-ij})$ is of class $\mathcal{C}^1$ (see Appendix A.2 for a review of differential calculus).

[7] For every $i \in I$, every $v_i \in \mathcal{D}_i$ and every $j \neq i$, the supremum in the definitions of $\|v_i\|_\infty$ and $\|\partial_{ij} v_i\|_\infty$ is attained, meaning that sup can be replaced by max. This follows from the fact that both v_i and $\partial_{ij} v_i$ are continuous functions and that $\mathbf{G}$ is a compact subspace of $\mathbb{R}^{\ell}$.

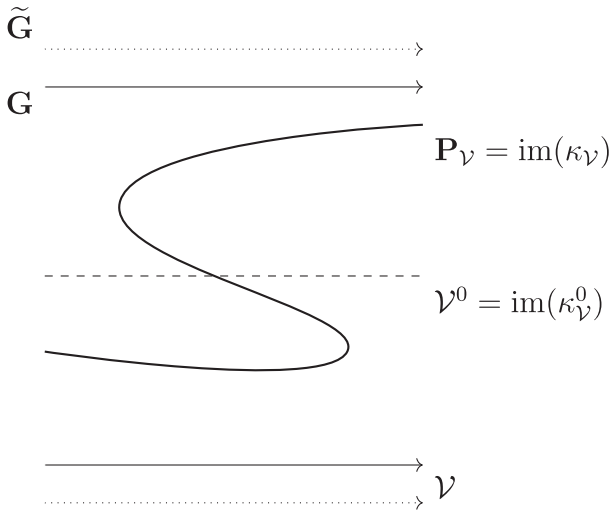

Fig. 1. $\mathbf{P}_{\mathcal{V}} = \mathrm{im}(\kappa_{\mathcal{V}})$ *(in thick line) and* $\mathcal{V}^0 = \mathrm{im}(\kappa^0_{\mathcal{V}})$ *(in dashed line).*

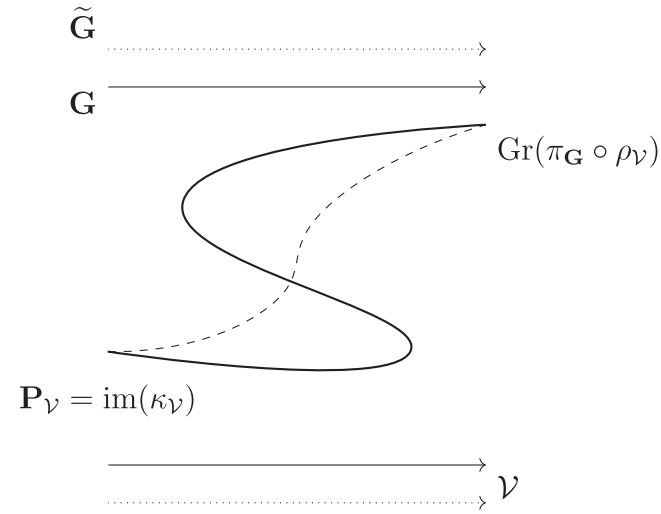

Fig. 2. *Deformation of* $\mathbf{P}_{\mathcal{V}}$ *(in thick line) into* $\mathrm{Gr}(\pi_{\mathbf{G}} \circ \rho_{\mathcal{V}})$ *(in dashed line) within the ambient space* $\widetilde{\mathbf{E}}_{\mathcal{V}}$.

In analogy to knot theory, which studies topological embeddings of the unit circle $\mathbb{S}^1 \subset \mathbb{R}^2$ into $\mathbb{R}^3$ or $\mathbb{S}^3$ (called knots), the term "unknot" in the previous definition is inspired by the trivial knot $(x, y) \in \mathbb{S}^1 \mapsto (x, y, 0) \in \mathbb{R}^3$, which is referred to as the unknot. Here, the idea is that for every set $\mathcal{V}$ of societies, the set of all topological embeddings of $\mathcal{V}$ into the ambient space $\widetilde{\mathbf{E}}_{\mathcal{V}}$ includes the trivial embedding, which is represented by $\kappa^0_{\mathcal{V}}$. This embedding maps $\mathcal{V}$ into $\widetilde{\mathbf{E}}_{\mathcal{V}}$ in a straightforward, non-interacting manner. Fig. 1 provides a simple representation of both the unknot $\kappa^0_{\mathcal{V}}$ associated with $\mathcal{V}$ and the knot $\kappa_{\mathcal{V}}$ of pairwise stable networks associated with $\mathcal{V}$, where $\mathcal{V} \subset \mathcal{S}$ is regular and $\rho_{\mathcal{V}}$ is the inverse of the homeomorphism $\eta_{\mathcal{V}}$ from Theorem 4.1.

Recall that for every topological spaces X, Y and every topological embeddings $e^1, e^2 : X \to Y$, an ambient isotopy between e^1 and e^2 is a continuous map $\theta : [0,1] \times Y \to Y$ such that: $\theta(0, \cdot) = \mathrm{id}_Y$, $\theta(1, \cdot) \circ e^1 = e^2$, and, for every $t \in [0,1]$, $\theta(t, \cdot)$ is a homeomorphism (see Appendix A.1 for more details). In light of Theorem 4.1, the following first main result of this paper sheds further light on the topological structure of the graph of pairwise stable networks associated with any regular set of societies.

Unknottedness theorem. *For every regular set $\mathcal{V}$ of societies, the knot $\kappa_{\mathcal{V}} : \mathcal{V} \to \widetilde{\mathbf{E}}_{\mathcal{V}}$ of pairwise stable networks associated with $\mathcal{V}$ is ambient isotopic to the unknot $\kappa^0_{\mathcal{V}} : \mathcal{V} \to \widetilde{\mathbf{E}}_{\mathcal{V}}$ associated with $\mathcal{V}$ within the ambient space $\widetilde{\mathbf{E}}_{\mathcal{V}}$, through an ambient isotopy $\theta_{\mathcal{V}}$ that does not deform the boundary of $\widetilde{\mathbf{E}}_{\mathcal{V}}$.*

The proof is provided in Appendix B. Consider a regular set $\mathcal{V}$ of societies. By analogy to knot theory, the term "unknottedness" refers to the fact that $\kappa_{\mathcal{V}} : \mathcal{V} \to \widetilde{\mathbf{E}}_{\mathcal{V}}$ can be seen as "equivalent" to the unknot $\kappa^0_{\mathcal{V}} : \mathcal{V} \to \widetilde{\mathbf{E}}_{\mathcal{V}}$ associated with $\mathcal{V}$. Intuitively, the unknottedness theorem asserts that the graph $\mathbf{P}_{\mathcal{V}} = \mathrm{im}(\kappa_{\mathcal{V}})$ of pairwise stable networks associated with $\mathcal{V}$ can be continuously deformed (using $\theta_{\mathcal{V}}$) into a trivial copy $\mathcal{V}^0 = \mathrm{im}(\kappa^0_{\mathcal{V}})$ of $\mathcal{V}$, within the ambient space $\widetilde{\mathbf{E}}_{\mathcal{V}}$. The key idea is that $\theta_{\mathcal{V}}$ does not deform only $\mathbf{P}_{\mathcal{V}}$, but the entire space $\widetilde{\mathbf{E}}_{\mathcal{V}}$.[8]

Note that the unknottedness theorem is stronger than Theorem 4.1: since $\kappa_{\mathcal{V}} : \mathcal{V} \to \widetilde{\mathbf{E}}_{\mathcal{V}}$ is ambient isotopic to $\kappa^0_{\mathcal{V}} : \mathcal{V} \to \widetilde{\mathbf{E}}_{\mathcal{V}}$ within $\widetilde{\mathbf{E}}_{\mathcal{V}}$, the images $\kappa_{\mathcal{V}}(\mathcal{V}) = \mathbf{P}_{\mathcal{V}}$ and $\kappa^0_{\mathcal{V}}(\mathcal{V}) = \mathcal{V}^0$ are homeomorphic, where $\mathcal{V}^0$ is itself homeomorphic to $\mathcal{V}$.

[8] In knot theory, the notion of homeomorphism alone is insufficient to classify knots since the image of $\mathbb{S}^1$ under any knot is homeomorphic to $\mathbb{S}^1$ itself. The (stronger) notion of ambient isotopy is more useful since it allows for a finer distinction between different knots.

Before providing a sketch of the proof of the unknottedness theorem, let us revisit two important works in game theory mentioned in the introduction: Demichelis and Germano (2002) and Predtetchinski (2009). These works are closely related to the results in this paper. Specifically, Demichelis and Germano established an unknottedness theorem for mixed-strategy Nash equilibria in finite strategic-form games. Predtetchinski, in addition to his topological structure theorem, generalized Demichelis–Germano's unknottedness theorem to Nash equilibria of own-strategy C^1 concave games. Furthermore, in the same paper, Demichelis and Germano developed several crucial results regarding the concept of Nash dynamics. These insights will be highly relevant in the upcoming section of this paper.

Sketch of proof. Let $\mathcal{V}$ be a fixed regular set of societies.

In Step I, we first show that the knot $\kappa_{\mathcal{V}} : \mathcal{V} \to \widetilde{\mathbf{E}}_{\mathcal{V}}$ of pairwise stable networks associated with $\mathcal{V}$ and the topological embedding $e^{\pi_{\mathbf{G}} \circ \rho_{\mathcal{V}}} : v \in \mathcal{V} \mapsto \big(v, (\pi_{\mathbf{G}} \circ \rho_{\mathcal{V}})(v)\big) \in \widetilde{\mathbf{E}}_{\mathcal{V}}$ are ambient isotopic within $\widetilde{\mathbf{E}}_{\mathcal{V}}$, through an ambient isotopy $\overline{\theta}^1_{\mathcal{V}}$. The idea is to continuously deform the graph $\mathbf{P}_{\mathcal{V}}$ of pairwise stable networks associated with $\mathcal{V}$ into the graph of the continuous map $\pi_{\mathbf{G}} \circ \rho_{\mathcal{V}} : \mathcal{V} \to \mathbf{G}$, within the ambient space. Fig. 2 illustrates this step.

In Step II, for every continuous map $f : \mathcal{V} \to \mathbf{G}$, we show that the topological embedding $e^f : v \in \mathcal{V} \mapsto (v, f(v)) \in \widetilde{\mathbf{E}}_{\mathcal{V}}$ and the unknot $\kappa^0_{\mathcal{V}} : \mathcal{V} \to \widetilde{\mathbf{E}}_{\mathcal{V}}$ associated with $\mathcal{V}$ are ambient isotopic within $\widetilde{\mathbf{E}}_{\mathcal{V}}$, through an ambient isotopy $\overline{\theta}^{2,f}_{\mathcal{V}}$. The goal is to continuously deform the graph of f into the trivial copy $\mathcal{V}^0$ of $\mathcal{V}$. Fig. 3 illustrates this step.

Finally, in Step III, using the ambient isotopies $\overline{\theta}^1_{\mathcal{V}}$ from Step I and $\overline{\theta}^{2,f}_{\mathcal{V}}$ from Step II (for $f = \pi_{\mathbf{G}} \circ \rho_{\mathcal{V}}$), we construct an ambient isotopy $\theta_{\mathcal{V}}$ between $\kappa_{\mathcal{V}}$ and $\kappa^0_{\mathcal{V}}$ within $\widetilde{\mathbf{E}}_{\mathcal{V}}$, completing the proof of the theorem. A key feature of $\theta_{\mathcal{V}}$ is that it does not deform the boundary of $\widetilde{\mathbf{E}}_{\mathcal{V}}$, which will be important in proving the dynamics equivalence theorem in Section 6.1.

6. Network dynamics

In this section, we introduce the notions of network dynamics and extended network dynamics on an arbitrary set $\mathcal{V} \subset \mathcal{S}$ of societies. Briefly, a network dynamics (resp. an extended network dynamics) on $\mathcal{V}$ is a family of vector fields $(D_v)_{v \in \mathcal{V}}$ on $\mathbf{G}$ (resp. $(\widetilde{D}_v)_{v \in \mathcal{V}}$ on $\widetilde{\mathbf{G}}$) such that for every $v \in \mathcal{V}$, the set of zeros of D_v (resp. of $\widetilde{D}_v$) coincides with the set of pairwise stable networks of v (see Appendix A.3 for some reminders about vector fields).

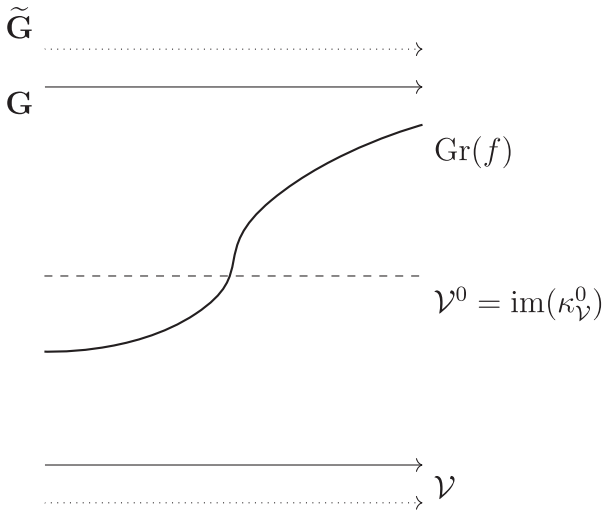

Fig. 3. *Deformation of* $\mathrm{Gr}(f)$ *(in thick line) into* $\mathcal{V}^0$ *(in dashed line) within the ambient space* $\widetilde{\mathbf{E}}_{\mathcal{V}}$.

Definition 6.1. Let $\mathcal{V} \subset \mathcal{S}$ be a set of societies:

- A *network dynamics on* $\mathcal{V}$ is a continuous map $D : \mathbf{E}_{\mathcal{V}} \to \mathrm{T}\mathbf{G}$ such that the following two conditions hold:
  1. For every $v \in \mathcal{V}$, $D_v = D(v, \cdot)$ is a vector field on $\mathbf{G}$.
  2. $\mathfrak{Z}(D) = \mathbf{P}_{\mathcal{V}}$, where $\mathfrak{Z}(D) = \{(v, g) \in \mathbf{E}_{\mathcal{V}} : g \in \mathfrak{Z}(D_v)\}$, with $\mathfrak{Z}(D_v)$ being the set of zeros of the vector field D_v.
- An *extended network dynamics on* $\mathcal{V}$ is a continuous map $\widetilde{D} : \widetilde{\mathbf{E}}_{\mathcal{V}} \to \mathrm{T}\widetilde{\mathbf{G}}$ such that the following two conditions hold:
  1. For every $v \in \mathcal{V}$, $\widetilde{D}_v = \widetilde{D}(v, \cdot)$ is a vector field on $\widetilde{\mathbf{G}}$.
  2. $\mathfrak{Z}(\widetilde{D}) = \mathbf{P}_{\mathcal{V}}$, where $\mathfrak{Z}(\widetilde{D}) = \{(v, g) \in \widetilde{\mathbf{E}}_{\mathcal{V}} : g \in \mathfrak{Z}(\widetilde{D}_v)\}$, with $\mathfrak{Z}(\widetilde{D}_v)$ being the set of zeros of the vector field $\widetilde{D}_v$.

Remark 6.1 (**TG** *and* T$\widetilde{\mathbf{G}}$ *are Trivial*)**.** Note that $\mathbf{G}$ (resp. $\widetilde{\mathbf{G}}$) is a convex subset of $\mathbb{R}^{\ell}$, which implies that it is contractible in $\mathbb{R}^{\ell}$, thus the tangent bundle T$\mathbf{G}$ of $\mathbf{G}$ (resp. T$\widetilde{\mathbf{G}}$ of $\widetilde{\mathbf{G}}$) is trivial (see Theorem A.1).

Now, consider the following results (see Appendix A.1 for some reminders about vector bundles).

Proposition 6.1. *Let M be a smooth n-manifold with a trivial tangent bundle, and let X be a topological space. For every vector bundle isomorphism $\zeta : \mathrm{T}M \to M \times \mathbb{R}^n$, there exists a bijection Ψ_ζ from $C^0(X \times M, \mathbb{R}^n)$ to*

$$\mathrm{Dyn}(X, M) = \{D \in C^0(X \times M, \mathrm{T}M) : \forall x \in X, D(x, \cdot) \in \mathfrak{X}(M)\}$$

with the following properties:

1. *For every $f, f' \in C^0(X \times M, \mathbb{R}^n)$, there exists a homotopy between f and f' if and only if there exists a homotopy H between $\Psi_\zeta(f)$ and $\Psi_\zeta(f')$ such that for every $t \in [0, 1]$, $H(t, \cdot, \cdot) \in \mathrm{Dyn}(X, M)$.*
2. *For every $f \in C^0(X \times M, \mathbb{R}^n)$, $f^{-1}(0_{\mathbb{R}^n}) = \{(x, a) \in X \times M : a \in \mathfrak{Z}(\Psi_\zeta(f)(x, \cdot))\}$.*

Corollary 6.1. *Let M be a smooth n-manifold with a trivial tangent bundle. For every vector bundle isomorphism $\zeta : \mathrm{T}M \to M \times \mathbb{R}^n$, there exists a bijection Ψ_ζ from $C^0(M, \mathbb{R}^n)$ to $\mathfrak{X}(M)$ with the following properties:*

1. *For every $f, f' \in C^0(M, \mathbb{R}^n)$, there exists a homotopy between f and f' if and only if there exists a homotopy H between $\Psi_\zeta(f)$ and $\Psi_\zeta(f')$ such that for every $t \in [0, 1]$, $H(t, \cdot) \in \mathfrak{X}(M)$.*
2. *For every $f \in C^0(M, \mathbb{R}^n)$, $f^{-1}(0_{\mathbb{R}^n}) = \mathfrak{Z}(\Psi_\zeta(f))$.*

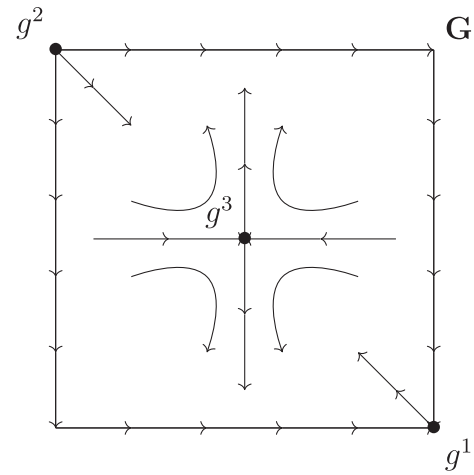

Fig. 4. *"Slice" of D, with* $\mathfrak{Z}(D_v) = \{g^1, g^2, g^3\}$.

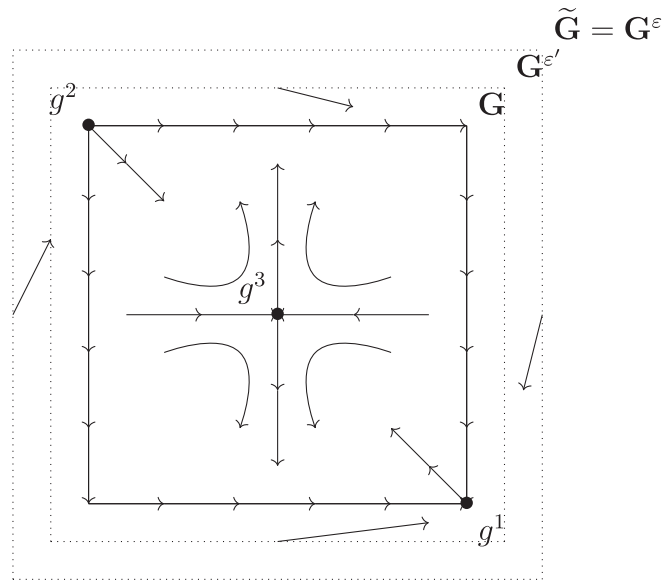

Fig. 5. *"Slice" of $\widetilde{D}$, with* $\mathfrak{Z}(\widetilde{D}_v) = \{g^1, g^2, g^3\}$.

The proofs are provided in Appendix C. According to Remark 6.1, Proposition 6.1 states that any network dynamics D on a set $\mathcal{V} \subset \mathcal{S}$ of societies can be treated as a continuous map from $\mathbf{E}_{\mathcal{V}}$ to $\mathbb{R}^{\ell}$ such that $D^{-1}(0_{\mathbb{R}^{\ell}}) = \mathbf{P}_{\mathcal{V}}$. Additionally, any extended network dynamics $\widetilde{D}$ on $\mathcal{V}$ can be treated as a continuous map from $\widetilde{\mathbf{E}}_{\mathcal{V}}$ to $\mathbb{R}^{\ell}$ such that $\widetilde{D}^{-1}(0_{\mathbb{R}^{\ell}}) = \mathbf{P}_{\mathcal{V}}$. This identification may sometimes be implicitly or explicitly used in some definitions or proofs to avoid complex details.

In the following, we introduce the notions of an inward-pointing network dynamics and a strongly inward-pointing extended network dynamics. Before doing so, recall that for every convex subset C of $\mathbb{R}^n$ and every $x \in C$, the normal cone to C at x is defined as

$$\mathrm{N}_C(x) = \{y \in \mathbb{R}^n : \forall x' \in C, \langle y, x' - x \rangle \leq 0\}$$

(see, e.g., Rockafellar and Wets, 1998, pp. 203–204).

Definition 6.2. Let $\mathcal{V} \subset \mathcal{S}$ be a set of societies. A network dynamics D on $\mathcal{V}$ is said to be *inward-pointing* if for every $v \in \mathcal{V}$, every $g \in \partial\mathbf{G}$ and every $x \in \mathrm{N}_{\mathbf{G}}(g)$, we have $\langle D_v(g), x \rangle \leq 0$.

Fig. 4 illustrates a "slice" of an inward-pointing network dynamics D for $\ell = 2$, for some set $\mathcal{V} \subset \mathcal{S}$ of societies, and some $v \in \mathcal{V}$.

Definition 6.3. Let $\mathcal{V} \subset \mathcal{S}$ be a set of societies. An extended network dynamics $\widetilde{D}$ on $\mathcal{V}$ is said to be *strongly inward-pointing* if for every $v \in \mathcal{V}$, every $\varepsilon' \in (0, \varepsilon]$, every $g \in \partial \mathbf{G}^{\varepsilon'}$ and every $x \in \mathrm{N}_{\mathbf{G}^{\varepsilon'}}(g) \backslash \{0_{\mathbb{R}^\ell}\}$, we have $\langle \widetilde{D}_v(g), x \rangle < 0$.

Similarly to the previous case, Fig. 5 provides a "slice" of a strongly inward-pointing extended network dynamics $\widetilde{D}$ for $\ell = 2$, for some set $\mathcal{V} \subset \mathcal{F}$ of societies, and some $v \in \mathcal{V}$. The two notions of inward-pointing network dynamics and of strongly inward-pointing extended network dynamics are in fact quite natural. If one wants to deal with flows associated with vector fields on $\mathbf{G}$, then the first notion is worth to consider; at the boundary of $\mathbf{G}$, we do not want to move "outside". The second notion seeks to specify how an extended network dynamics should "normally" behave on the ambient space $\widetilde{\mathbf{E}}_{\mathcal{V}}$ (where $\mathcal{V} \subset \mathcal{S}$), which corresponds to an "artificial enlargement" of the space $\mathbf{E}_{\mathcal{V}}$. Indeed, the practical feature of $\widetilde{\mathbf{E}}_{\mathcal{V}}$ is that it allows us to describe the behavior outside $\mathbf{G}$ (especially around isolated zeros on $\partial \mathbf{G}$). However, starting from any element in $\widetilde{\mathbf{G}} \backslash \mathbf{G}$, we want to ensure that the flow moves back to $\mathbf{G}$ (following the path induced by the underlying extended network dynamics). This will be guaranteed if the considered extended network dynamics is strongly inward-pointing.

Now, before presenting the remaining results of this paper, we present an example of inward-pointing network dynamics, which is analogous to the best-reply dynamics in game theory.

Example 6.1 (*Pairwise Best-Response Dynamics*)**.** For every $i \in I$, define

$$\mathcal{V}_i = \{v_i \in C^0(X, \mathbb{R}) : \forall j \neq i, \forall g_{-ij} \in \mathbf{G}_{-ij}, v_i(\cdot, g_{-ij}) \text{ is strictly concave}\},$$

and let $\mathcal{V} = \prod_{i \in I} \mathcal{V}_i$. Consider the map

$$D : (v, g) \in \mathcal{V} \times \mathbf{G} \mapsto \left(\min\{w_i(v_i, g_{-ij}), w_j(v_j, g_{-ij})\} - g_{ij}\right)_{ij \in \mathbf{L}} \in \mathbb{R}^\ell,$$

where for every $(v, g) \in \mathcal{V} \times \mathbf{G}$, and every $ij \in \mathbf{L}$, $w_i(v_i, g_{-ij})$ is the (unique) argument that maximizes $v_i(\cdot, g_{-ij})$ over $[0, 1]$, and similarly, $w_j(v_j, g_{-ij})$ maximizes $v_j(\cdot, g_{-ij})$. We show that D is an inward-pointing network dynamics.

First, observe that $D^{-1}(0_{\mathbb{R}^\ell}) = \mathbf{P}_{\mathcal{V}}$. Indeed, for every $(v, g) \in \mathcal{V} \times \mathbf{G}$, g is pairwise stable with respect to v if and only if for every $ij \in \mathbf{L}$, we have $g_{ij} = \min\{w_i(v_i, g_{-ij}), w_j(v_j, g_{-ij})\}$ (this follows from the strict concavity of the maps $v_i(\cdot, g_{-ij})$ and $v_j(\cdot, g_{-ij})$). Equivalently, g is pairwise stable with respect to v if and only if for every $ij \in \mathbf{L}$, $\min\{w_i(v_i, g_{-ij}), w_j(v_j, g_{-ij})\} - g_{ij} = 0$, i.e., if and only if $D(v, g) = 0_{\mathbb{R}^\ell}$.

Next, we show that D is inward-pointing. Let $v \in \mathcal{V}$, $g \in \partial \mathbf{G}$, and $x \in \mathrm{N}_{\mathbf{G}}(g)$. By definition, $\langle \gamma - g, x \rangle \leq 0$, for every $\gamma \in \mathbf{G}$. In particular, note that $\langle \hat{\gamma} - g, x \rangle \leq 0$, where $\hat{\gamma} = \left(\min\{w_i(v_i, g_{-ij}), w_j(v_j, g_{-ij})\}\right)_{ij \in \mathbf{L}} \in \mathbf{G}$. Finally, observe that

$$D_v(g) = \left(\min\{w_i(v_i, g_{-ij}), w_j(v_j, g_{-ij})\} - g_{ij}\right)_{ij \in \mathbf{L}} = \left(\hat{\gamma}_{ij} - g_{ij}\right)_{ij \in \mathbf{L}} = \hat{\gamma} - g.$$

Thus, $\langle D_v(g), x \rangle \leq 0$.

In game theory, the best-reply dynamics describes a process where players iteratively adjust their strategies to maximize their payoffs given the current strategies of their opponents. This dynamics often leads to convergence toward a Nash equilibrium or, in some cases, to cyclical behavior. Similarly, the pairwise best-response dynamics follows a comparable adjustment process, but applied to the formation of weighted networks. In this setting:

- Each agent i evaluates potential link weights based on a strictly concave payoff function $v_i(\cdot, g_{-ij})$, ensuring the existence of a unique optimal choice.
- The actual adjustment of the weight g_{ij} is determined by the minimum of the two agents' optimal choices (according to the definition of pairwise stability).

The mechanism closely resembles the best-reply dynamics:

1. Each pair of agents updates its decision (link strength) based on their optimal response to the current network state.
2. This network dynamics naturally guides the system toward a stable configuration where no agent has an incentive to deviate, akin to how best-reply dynamics tends to lead to a Nash equilibrium in standard games.

One key distinction is that, in this network setting, the "strategy space" consists of continuous link weights rather than discrete actions, and adjustments are constrained by mutual agreement (i.e., the minimum function). This ensures that the network evolves smoothly under a structured and well-behaved optimization process, much like in continuous best-reply dynamics in population games or learning dynamics in economic networks. Thus, the pairwise best-response dynamics can be interpreted as a best-reply-like process where agents continuously adjust their links to maximize their individual utilities, leading to stable network configurations in a way analogous to Nash equilibrium selection in standard game dynamics.

To tie the discussion in the introduction, let us compare, for instance, the pairwise best-response dynamics with the dynamic process in Watts (2001):

1. The pairwise best-response dynamics models weighted networks, where link formation is not binary but instead involves continuous adjustments of link strengths. Watts, on the other hand, considers unweighted networks, where links are either present or absent: agents decide to form or remove links based on a cost–benefit analysis, leading to discrete changes in network structure.
2. The pairwise best-response dynamics is deterministic: link weights evolve according to a structured rule that depends on the best responses of the two agents involved. Each weight is adjusted to the minimum of the agents' optimal choices, ensuring a smooth and predictable evolution of the network. In contrast, Watts studies a stochastic network formation process, where link adjustments occur probabilistically. Specifically, agents consider forming or severing links based on a random mutation process (akin to stochastic stability in evolutionary game theory). This introduces an element of noise and randomness that affects the long-term structure of the network.
3. The pairwise best-response dynamics is structured around a best-reply-like mechanism: agents choose their optimal link weights based on their strictly concave payoff functions, ensuring that the adjustments follow a clear optimization process. In Watts' model, agents act more myopically, making local pairwise decisions about adding or removing links based on immediate gains. The long-term network structure emerges from repeated interactions rather than from an explicit optimization process.

To summarize, the pairwise best-response dynamics offers a continuous, deterministic, and optimization-based approach to network evolution, whereas Watts adopts a discrete, stochastic, and evolutionary framework. While Watts' model is well-suited for understanding long-run network emergence under uncertainty, our approach provides a structured way to study equilibrium behavior in weighted networks.

6.1. The dynamics equivalence theorem

The second main result of this paper, called the dynamics equivalence theorem, states that any two strongly inward-pointing extended network dynamics on an arbitrary regular set $\mathcal{V}$ of societies are homotopic within the set of all extended network dynamics on $\mathcal{V}$.

Dynamics equivalence theorem. *For every regular set $\mathcal{V}$ of societies, any two strongly inward-pointing extended network dynamics on $\mathcal{V}$ are homotopic, through a homotopy $\widetilde{H}_{\mathcal{V}}$ such that for every $t \in [0, 1]$, $\widetilde{H}_{\mathcal{V}}(t, \cdot, \cdot)$ is an extended network dynamics on $\mathcal{V}$.*

The proof is provided in Appendix D.

As in Section 5, before providing a sketch of the proof of the dynamics equivalence theorem, let us recall the work of Demichelis and Germano on Nash dynamics (Demichelis and Germano, 2002), in the framework of mixed-strategy Nash equilibria in finite strategic-form games.

Sketch of proof. Consider a fixed regular set $\mathcal{V}$ of societies. Based on Remark 6.1 and on Proposition 6.1, we treat any extended network dynamics on $\mathcal{V}$ as a continuous map $\widetilde{D} : \widetilde{\mathbf{E}}_{\mathcal{V}} \to \mathbb{R}^{\ell}$ such that $\widetilde{D}^{-1}(0_{\mathbb{R}^{\ell}}) = \mathbf{P}_{\mathcal{V}}$, throughout this proof, in order to simplify the demonstration.

Now, consider two strongly inward-pointing extended network dynamics $\widetilde{D}, \widetilde{D}' : \widetilde{\mathbf{E}}_{\mathcal{V}} \to \mathbb{R}^{\ell}$ on $\mathcal{V}$. The required homotopy $\widetilde{H}_{\mathcal{V}}$ between $\widetilde{D}$ and $\widetilde{D}'$ is built in a two-step process:

$$\begin{cases} \widetilde{H}_{\mathcal{V}} : & [0,1]\times\widetilde{\mathbf{E}}_{\mathcal{V}} \to \mathbb{R}^{\ell} \\ & (t,(v,g)) \mapsto \begin{cases} \widetilde{H}^1_{\mathcal{V}}(2t,(v,g)) & \text{if} \quad t\in[0,\frac{1}{2}] \\ \widetilde{H}^2_{\mathcal{V}}(2t-1,(v,g)) & \text{if} \quad t\in[\frac{1}{2},1] \end{cases} \end{cases},$$

The maps $\widetilde{H}^1_{\mathcal{V}}$ and $\widetilde{H}^2_{\mathcal{V}}$ are explained below.

In Step I, we construct the map $\widetilde{H}^1_{\mathcal{V}}$. We deform the map $\widetilde{D}$ into a continuous map $\widetilde{D}_{\partial} : \widetilde{\mathbf{E}}_{\mathcal{V}} \to \mathbb{R}^{\ell}$ that is equal to $\widetilde{D}$ on $\mathbf{E}_{\mathcal{V}}$, and which is equal to $\widetilde{D}'$ on $\mathcal{V}\times\partial\widetilde{\mathbf{G}}$, i.e.,

$$\widetilde{D}_{\partial}|_{\mathbf{E}_{\mathcal{V}}} = \widetilde{D}|_{\mathbf{E}_{\mathcal{V}}} \text{ and } \widetilde{D}_{\partial}|_{\mathcal{V}\times\partial\widetilde{\mathbf{G}}} = \widetilde{D}'|_{\mathcal{V}\times\partial\widetilde{\mathbf{G}}}.$$

This property is crucial for the construction of $\widetilde{H}^2_{\mathcal{V}}$. We define $\widetilde{H}^1_{\mathcal{V}}$ as the straight-line homotopy between $\widetilde{D}$ and $\widetilde{D}_{\partial}$. For every $t\in[0,1]$, $\widetilde{H}^1_{\mathcal{V}}(t,\cdot,\cdot)$ is an extended network dynamics on $\mathcal{V}$.

From Step II to Step IV, we construct the map $\widetilde{H}^2_{\mathcal{V}}$. We aim to deform the map $\widetilde{D}_{\partial}$ into the map $\widetilde{D}'$. To simplify this process, we use the ambient isotopy $\theta_{\mathcal{V}} : [0,1]\times\widetilde{\mathbf{E}}_{\mathcal{V}} \to \widetilde{\mathbf{E}}_{\mathcal{V}}$ of the unknottedness theorem (Step II). Instead of directly deforming $\widetilde{D}_{\partial}$ into $\widetilde{D}'$, we deform the map

$$\widetilde{\Delta}_{\partial} = \widetilde{D}_{\partial}\circ(\theta_{\mathcal{V}}(1,\cdot,\cdot))^{-1}$$

into the map

$$\widetilde{\Delta}' = \widetilde{D}'\circ(\theta_{\mathcal{V}}(1,\cdot,\cdot))^{-1}.$$

By finding a homotopy $\widetilde{\boldsymbol{\Phi}}_{\mathcal{V}}$ between $\widetilde{\Delta}_{\partial}$ and $\widetilde{\Delta}'$ such that for every $t\in[0,1]$, $(\widetilde{\boldsymbol{\Phi}}_{\mathcal{V}}(t,\cdot,\cdot))^{-1}(0_{\mathbb{R}^{\ell}}) = \mathcal{V}^0$, we can define $\widetilde{H}^2_{\mathcal{V}}$ as:

$$\begin{cases} \widetilde{H}^2_{\mathcal{V}} : & [0,1]\times\widetilde{\mathbf{E}}_{\mathcal{V}} \to \mathbb{R}^{\ell} \\ & (t,(v,g)) \mapsto (\widetilde{\boldsymbol{\Phi}}_{\mathcal{V}}(t,\cdot,\cdot)\circ\theta_{\mathcal{V}}(1,\cdot,\cdot))(v,g) \end{cases},$$

which is a homotopy between $\widetilde{D}_{\partial}$ and $\widetilde{D}'$, ensuring that for every $t\in[0,1]$, $\widetilde{H}^2_{\mathcal{V}}(t,\cdot,\cdot)$ is an extended network dynamics on $\mathcal{V}$. In Step III, we perform a "back-and-forth" motion in order to build the homotopy $\widetilde{\boldsymbol{\Phi}}_{\mathcal{V}}$ between $\widetilde{\Delta}_{\partial}$ and $\widetilde{\Delta}'$, as mentioned above. The following diagram summarizes this construction:

$$\begin{tikzcd} \widetilde{\Delta}_{\partial} \arrow[r, "\widetilde{\Phi}^1_{\mathcal{V}}"] \arrow[d, dashed] & \widetilde{\Delta}^1 \arrow[r, "\widetilde{\Phi}^2_{\mathcal{V}}"] & \widehat{\Delta}^2 \arrow[d, equal] \\ \widetilde{\Delta}' & \widehat{\Delta}^1 \arrow[l, "\widetilde{\Phi}^4_{\mathcal{V}}"] & \widehat{\Delta}^2 \arrow[l, "\widetilde{\Phi}^3_{\mathcal{V}}"] \end{tikzcd}$$

Each map of this diagram is detailed in the proof. However, the key point is the equality in the last column of the diagram between

$$\begin{cases} \widetilde{\Delta}^2 : & \widetilde{\mathbf{E}}_{\mathcal{V}} \to \mathbb{R}^{\ell} \\ & (v,g) \mapsto \begin{cases} \frac{\widetilde{\Delta}_{\partial}(v,\gamma^g_{\partial})}{\|\widetilde{\Delta}_{\partial}(v,\gamma^g_{\partial})\|_2}\|g-g^0\|_2 & \text{if} \quad g\neq g^0 \\ 0_{\mathbb{R}^{\ell}} & \text{otherwise} \end{cases} \end{cases}$$

and

$$\begin{cases} \widehat{\Delta}^2 : & \widetilde{\mathbf{E}}_{\mathcal{V}} \to \mathbb{R}^{\ell} \\ & (v,g) \mapsto \begin{cases} \frac{\widetilde{\Delta}'(v,\gamma^g_{\partial})}{\|\widetilde{\Delta}'(v,\gamma^g_{\partial})\|_2}\|g-g^0\|_2 & \text{if} \quad g\neq g^0 \\ 0_{\mathbb{R}^{\ell}} & \text{otherwise} \end{cases} \end{cases},$$

where for every $g\in\widetilde{\mathbf{G}}\backslash\{g^0\}$, $\gamma^g_{\partial}\in\partial\widetilde{\mathbf{G}}$ is detailed in Substep III.1. This equality holds because:

1. The ambient isotopy $\theta_{\mathcal{V}}$ does not deform the boundary of the ambient space $\widetilde{\mathbf{E}}_{\mathcal{V}}$.
2. The map $\widetilde{H}^1_{\mathcal{V}}$ (built in Step I) ensures that the homotopy $\widetilde{H}^2_{\mathcal{V}}$ behaves as required.

Finally, the map $\widetilde{\boldsymbol{\Phi}}_{\mathcal{V}}$ is constructed using the homotopies $(\widetilde{\boldsymbol{\Phi}}^p_{\mathcal{V}})^4_{p=1}$ (details of these maps are provided in the proof). We conclude the construction of $\widetilde{H}^2_{\mathcal{V}}$ and $\widetilde{H}_{\mathcal{V}}$ in Step IV.

6.2. The indices equality theorem

Recall that for a given vector field X on an arbitrary smooth manifold, and a given isolated zero z of X, the index of X at z is a topological invariant that helps quantify the behavior of X near z The index characterizes the nature of X around z — whether it behaves like a source, a sink, a saddle, or has another type of singularity. The formal definition of the index is provided in Appendix A.3.

The following corollary, which stems from the dynamics equivalence theorem and constitutes the third main result of this paper, establishes a connection between the indices of strongly inward-pointing extended network dynamics on $\mathcal{V}$ at any isolated zero, i.e., at any isolated pairwise stable network (here, $\mathcal{V}$ represents an arbitrary regular set of societies).

Indices equality theorem. *Let $\mathcal{V}$ be a regular set of societies. For every strongly inward-pointing extended network dynamics $\widetilde{D}$ and $\widetilde{D}'$ on $\mathcal{V}$, and for every $v\in\mathcal{V}$, the index of $\widetilde{D}_v$ at g is equal to the index of $\widetilde{D}'_v$ at g, for every isolated point g of $\mathfrak{Z}(\widetilde{D}_v) = \mathfrak{Z}(\widetilde{D}'_v)$.*

The proof is provided in Appendix E.

Sketch of proof. Consider a fixed regular set $\mathcal{V}$ of societies and a fixed society $v\in\mathcal{V}$. Throughout this proof, we will treat any extended network dynamics on $\mathcal{V}$ as defined in Definition 6.1, without taking into account Remark 6.1 and Proposition 6.1.

Now, consider two strongly inward-pointing extended network dynamics $\widetilde{D}, \widetilde{D}' : \widetilde{\mathbf{E}}_{\mathcal{V}} \to \mathrm{T}\widetilde{\mathbf{G}}$ on $\mathcal{V}$. We aim to prove that the two maps $[\Psi_{\mathfrak{c}}^{-1}(\phi_*\widetilde{D}_v)]^{\phi(g)} : \partial\overline{B}(\phi(g),\epsilon) \to \mathbb{S}^{\ell-1}$ and $[\Psi_{\mathfrak{c}}^{-1}(\phi_*\widetilde{D}'_v)]^{\phi(g)} : \partial\overline{B}(\phi(g),\epsilon) \to \mathbb{S}^{\ell-1}$, for some isolated zero g of $\widetilde{D}_v$ (which is also isolated for $\widetilde{D}'_v$), some interior chart (U,ϕ) of $\widetilde{\mathbf{G}}$ containing g and some particular real number $\epsilon > 0$, are homotopic (see Appendix A.3 for some useful definitions in differential geometry).[9] Once this is done, we will be able to use the homotopy property of topological degree (see Proposition A.1) in order to deduce the result.

In Step I, we construct a homotopy $\widetilde{H}_v$ between $\widetilde{D}_v$ and $\widetilde{D}'_v$ using the homotopy $\widetilde{H}_{\mathcal{V}}$ from the dynamics equivalence theorem.

In Step II, we use the definition from Appendix A.3 to compute the index of $\widetilde{D}_v$ at the isolated zero g, which corresponds to the topological degree of the map $[\Psi_{\mathfrak{c}}^{-1}(\phi_*\widetilde{D}_v)]^{\phi(g)}$. We also show that the index of $\widetilde{D}'_v$ at g can be computed similarly using the chart (U,ϕ) and the real number ϵ.

In Step III, we explicitly construct a homotopy $\widetilde{H}^{\phi(g)}_v$ between the two maps $[\Psi_{\mathfrak{c}}^{-1}(\phi_*\widetilde{D}_v)]^{\phi(g)}$ and $[\Psi_{\mathfrak{c}}^{-1}(\phi_*\widetilde{D}'_v)]^{\phi(g)}$, using the homotopy $\widetilde{H}_v$ from Step I and the canonical bijection $\Psi_{\mathfrak{c}} : \mathcal{C}^0(\widetilde{\mathbf{G}},\mathbb{R}^{\ell}) \to \mathfrak{X}(\widetilde{\mathbf{G}})$ from Corollary 6.1. The homotopy $\widetilde{H}^{\phi(g)}_v$ is constructed from a first homotopy $\widetilde{H}^1_v$ between the pushforward $\phi_*\widetilde{D}_v$ of $\widetilde{D}_v$ by ϕ and the pushforward $\phi_*\widetilde{D}'_v$ of $\widetilde{D}'_v$ by ϕ, and a second homotopy $\widetilde{H}^2_v$ between $\Psi_{\mathfrak{c}}^{-1}(\phi_*\widetilde{D}_v)$ and $\Psi_{\mathfrak{c}}^{-1}(\phi_*\widetilde{D}'_v)$.

Last, in Step IV, we conclude the proof by applying the homotopy property of topological degree (Proposition A.1).

[9] The map $\Psi_{\zeta_{\mathfrak{c}}} = \Psi_{\mathfrak{c}} : \mathcal{C}^0(\widetilde{\mathbf{G}},\mathbb{R}^{\ell}) \to \mathfrak{X}(\widetilde{\mathbf{G}})$ is the canonical bijection of Corollary 6.1 considering the vector bundle isomorphism $\zeta_{\mathfrak{c}} : \mathrm{T}\widetilde{\mathbf{G}} \to \widetilde{\mathbf{G}}\times\mathbb{R}^{\ell}$ of Remark A.1.

The following diagram summarizes the constructions of this proof:

$$\begin{tikzcd}
\mathrm{T}\widetilde{\mathbf{G}} \arrow[r, "\mathrm{T}\phi"] & \mathrm{T}\phi(U) \arrow[r, "\zeta_{\mathbf{c}}"] & \phi(U)\times\mathbb{R}^{\ell} \arrow[r, "\mathrm{pr}_{\mathbb{R}^{\ell}}"] & \mathbb{R}^{\ell} \\
{[0,1]\times\widetilde{\mathbf{G}}} \arrow[u, "\widetilde{H}_v"] & {[0,1]\times\phi(U)} \arrow[l, "\mathrm{id}_{[0,1]}\times\phi^{-1}"] \arrow[u, "\widetilde{H}^1_v"] \arrow[urr, dashed, "\widetilde{H}^2_v"] & &
\end{tikzcd}$$

6.3. The semialgebraic indices equality theorem

In this section, we focus on sets of societies whose payoff functions take the following form:

$$v_i(g) = \sum_{k=1}^{\kappa} \alpha_{i,k}\sigma_k(g),$$

where $\kappa > 0$, $\alpha_{i,k} \in \mathbb{R}$ for each $k \in \{1,\dots,\kappa\}$ and σ_k is a semialgebraic map from $\mathbf{G}$ to $\mathbb{R}$.

Definition 6.4. For every finite family $\sigma = (\sigma_k)_{k=1}^{\kappa}$ of maps from $\mathbf{G}$ to $\mathbb{R}$ and every $i \in I$,

$$\begin{aligned}\mathcal{P}(\sigma)_i &= \{g \in \mathbf{G} \mapsto \sum_{k=1}^{\kappa} \alpha_{i,k}\sigma_k(g) \in \mathbb{R} : \forall k \in \{1,\dots,\kappa\}, \alpha_{i,k} \in \mathbb{R}\} \\ &= \mathrm{Span}(\{\sigma_k : k \in \{1,\dots,\kappa\}\})\end{aligned}$$

Moreover, we define

$$\mathcal{P}(\sigma) = \prod_{i\in I} \mathcal{P}(\sigma)_i.$$

For every finite family $\sigma = (\sigma_k)_{k=1}^{\kappa}$ of maps from $\mathbf{G}$ to $\mathbb{R}$ and every $i \in I$, $\mathcal{P}(\sigma)_i$ (equipped with the usual function addition and scalar multiplication) is, by definition, a finite-dimensional vector space:

- $\mathcal{P}(\sigma)_i$ is endowed with the unique topology that makes it a Hausdorff topological vector space. Under this topology, every linear map from $\mathcal{P}(\sigma)_i$ to any other topological vector space is continuous.
- The map

  $$\varphi_i : v_i \in \mathcal{P}(\sigma)_i \mapsto (\alpha_{i,k})_{k=1}^{\kappa} \in \mathbb{R}^{\kappa}$$

  is a linear isomorphism. Furthermore, by definition, the map

  $$\varphi : v \in \mathcal{P}(\sigma) \mapsto \times_{i\in N}\varphi_i(v_i) \in \mathbb{R}^{|I|\kappa}$$

  is also a linear isomorphism.

Notations. Consider a finite family $\sigma = (\sigma_k)_{k=1}^{\kappa}$ of maps from $\mathbf{G}$ to $\mathbb{R}$ and a set $\mathcal{V} \subset \mathcal{P}(\sigma)$ of societies.[10] The set $\varphi(\mathcal{V})$ of coefficients associated with societies in $\mathcal{V}$ is denoted by $\mathbf{C}_{\mathcal{V}}$. By abuse of notation, both the restriction of φ from $\mathcal{V}$ to $\mathbb{R}^{|I|\kappa}$ and the restriction of φ from $\mathcal{V}$ to $\mathbf{C}_{\mathcal{V}}$ are denoted by φ. However, note that $\varphi : \mathcal{V} \to \mathbf{C}_{\mathcal{V}}$ is a homeomorphism. For every $\alpha \in \mathbf{C}_{\mathcal{V}}$, the society in $\mathcal{V}$ whose coefficients correspond to α is denoted by v^{α}, i.e., $v^{\alpha} = \varphi^{-1}(\alpha)$.

We now introduce a stronger notion of regularity compared to the one in Definition 4.4.

Definition 6.5. A set $\mathcal{V} \subset \mathcal{S}$ of societies is said to be *semialgebraically regular* if the following four conditions hold:

1. **Concavity:** $\mathcal{V} \subset \mathcal{C}$.
2. **Differentiability:** $\mathcal{V} \subset \mathcal{D}$.
3. **$\mathcal{A}$-invariance:** $\mathcal{V} + \mathcal{A} = \mathcal{V}$.
4. **Semialgebraicity:** There exists a finite family $\sigma = (\sigma_k)_{k=1}^{\kappa}$ of semialgebraic maps from $\mathbf{G}$ to $\mathbb{R}$ such that $\mathcal{V} \subset \mathcal{P}(\sigma)$, and $\mathbf{C}_{\mathcal{V}}$ is a semialgebraic set.

[10] Note that if $\mathcal{V} \subset \mathcal{D}$, then the topology on $\mathcal{V}$ induced by $\mathcal{D}$ coincides with the one induced by $\mathcal{P}(\sigma)$.

Recall that a semialgebraic subset G of a semialgebraic set S is said to be generic in S (or a generic subset of S) if $\dim(S\backslash G) < \dim(S)$, and if G is open in S (see Appendix A.4 for a brief review about real semialgebraic geometry). Additionally, recall the following result concerning generic oddness of graphs of pairwise stable networks in the semialgebraic case.

Theorem 6.1 (Bich and Fixary, 2022). *For every semialgebraically regular set $\mathcal{V}$ of societies, there exists a generic semialgebraic subset G of $\mathbf{C}_{\mathcal{V}}$ such that for every $\alpha \in G$, the society v^{α} has an odd number of pairwise stable networks.*

Bich–Fixary's oddness theorem is originally established only in the case where $\sigma = (\sigma_k)_{k=1}^{\kappa}$ is an arbitrary family of polynomial functions. However, a similar argument shows that the result holds in the semialgebraic case.

Now, observe that within the framework of semialgebraically regular sets of societies, one can combine the indices equality theorem (see Section 6.2) with Theorem 6.1 to derive an interesting version variant of this result. This idea forms the core of the fourth and last main result of this paper.

Semialgebraic indices equality theorem. *Let $\mathcal{V}$ be a semialgebraically regular set of societies. There exists a generic semialgebraic subset G of $\mathbf{C}_{\mathcal{V}}$ such that for every strongly inward-pointing extended network dynamics $\widetilde{D}$ and $\widetilde{D}'$ on $\mathcal{V}$, and every $\alpha \in G$, the index of $\widetilde{D}_{v^{\alpha}}$ at g is equal to the index of $\widetilde{D}'_{v^{\alpha}}$ at g, for every point g of $\mathfrak{Z}(\widetilde{D}_{v^{\alpha}}) = \mathfrak{Z}(\widetilde{D}'_{v^{\alpha}})$.*

The proof follows directly from the ones of indices equality theorem and of Bich–Fixary's oddness theorem. The key difference between the indices equality theorem and the semialgebraic indices equality theorem lies in the fact that if we consider a semialgebraically regular set $\mathcal{V}$ of societies and strongly inward-pointing extended network dynamics $\widetilde{D}, \widetilde{D}'$ on $\mathcal{V}$, then for "almost every" society $v \in \mathcal{V}$, we can now meaningfully discuss the equality of the indices of $\widetilde{D}_v$ and $\widetilde{D}'_v$ at any zero, since in that case, each zero is, by definition, isolated.

Now, let us briefly discuss the potential applications of the previous result. The interesting point here does not lie on the construction of the network dynamics itself, but rather in the potential sets of societies that satisfy assumptions in Definition 6.5 (and in particular, the semialgebraicity assumption). In fact, as explained in Bich and Fixary (2022), many revisited weighted versions of models in network formation theory satisfy the aforementioned assumptions. For example, one can mention Bramoullé–Kranton's public good provision model (Bramoullé and Kranton, 2007), Calvó-Armengol–İlkılıç's information transmission model (Calvó-Armengol and İlkılıç, 2009), or Bala–Goyal's two-way flow model (Bala and Goyal, 2000). Therefore, one could construct network dynamics based on these models.

6.4. Index of a zero of a network dynamics

Definition 6.6. Let $\mathcal{V} \subset \mathcal{S}$ be a set of societies and D be an inward-pointing network dynamics on $\mathcal{V}$. A strongly inward-pointing extended network dynamics D^{ϵ} on $\mathcal{V}$ is called a *(ϵ-)extension of D to $\widetilde{\mathbf{E}}_{\mathcal{V}}$* if $D^{\epsilon}|_{\mathbf{E}_{\mathcal{V}}} = D$.

The following proposition establishes a connection between inward-pointing network dynamics and strongly inward-pointing extended network dynamics.

Proposition 6.2. *For every set $\mathcal{V} \subset \mathcal{S}$ of societies, any inward-pointing network dynamics on $\mathcal{V}$ admits an extension to $\widetilde{\mathbf{E}}_{\mathcal{V}}$.*

The proof is provided in Appendix F. Given a set $\mathcal{V}$ of societies, an inward-pointing network dynamics D on $\mathcal{V}$ and any extension D^{ϵ} of D to $\widetilde{\mathbf{E}}_{\mathcal{V}}$ (whose existence is guaranteed by Proposition 6.2), the indices equality theorem states that for every $v \in \mathcal{V}$ and every isolated zero g of D^{ϵ}_v, the index of D^{ϵ}_v at g is independent of the choice of

extension. Thus, even though we technically define the index of D_v at an isolated zero g on $\partial\mathbf{G}$ ($v \in \mathcal{V}$), we can still describe the behavior of D_v around g using any extension D^ϵ. Since such an extension is strongly inward-pointing, no additional zeros appear outside $\mathbf{G}$, making it a "faithful representation" of D on $\mathbf{E}^\epsilon_{\mathcal{V}}$. In particular, this highlights why the ambient space $\mathbf{E}^\epsilon_{\mathcal{V}}$ is useful to consider.

7. Conclusion

The initial goal of this paper was to contribute to the construction of a bridge linking game theory and network formation theory. To achieve this, we introduced four main results:

1. The *unknottedness theorem*, which provides topological results about graphs of pairwise stable networks (in the spirit of Demichelis–Germano's or Predtetchinski's results).
2. The *dynamics equivalence theorem*, which offers insights into the deformations of strongly inward-pointing extended network dynamics (in the spirit of Demichelis–Germano's result).
3. The *indices equality theorem*, which helps identify the behavior of any strongly inward-pointing extended network dynamics at any pairwise stable weighted network (still in the spirit of Demichelis–Germano's result).
4. The *semialgebraic indices equality theorem*, which establishes a link between the indices equality theorem and Bich–Fixary's semialgebraic oddness theorem.

Although the results of this paper may seem somewhat technical, they offer a new conceptual approach to the study of dynamic processes in network formation theory, for the reasons outlined in the introduction. In particular, network dynamics offer a fundamentally different perspective on dynamic formation processes. This is because, first, the networks considered here are weighted, and second, the continuous framework adopted in this paper allows, for instance, an interesting potential analysis of equilibria.

Starting from our study of graphs of pairwise stable networks and network dynamics, we believe that two main questions are worth exploring further. First, one could investigate whether the results related to the topological properties of graphs of pairwise stable networks (topological structure theorem and unknottedness theorem) could be extended to broader spaces of societies beyond those used in this paper (i.e., regular ones). Second, exploring the potential identification of the notions of index and degree of a pairwise stable weighted network — following an approach similar to Demichelis and Germano's, based on a result from Govindan and Wilson (1997) regarding Gül–Pearce—Stacchetti's Nash dynamics (Gul et al., 1993) – could be a promising research direction. Of course, from a more practical perspective, we believe that many interesting network dynamics could be built, either by adapting existing Nash dynamics (e.g., the replicator dynamics) or by studying more deeply specific types of network structures (e.g., networks of firms and potential alliances). The results presented in this paper may assist network theorists in characterizing the behavior of certain network dynamics around each equilibrium (i.e., each pairwise stable weighted network).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Mathematical reminders

A.1. Topology and vector bundles

A.1.1. General notions of topology

Definition A.1. Let X, Y be two topological spaces:

- A map $f : X \to Y$ is *proper* if for every compact subspace K of Y, $f^{-1}(K)$ is a compact subspace of X.
- A map $f : X \to Y$ is a *homeomorphism* if f is a bijection, and if f and f^{-1} are continuous maps. If such a map exists, then X and Y are said to be *homeomorphic*.
- A map $f : X \to Y$ is a *topological embedding* if the map $x \in X \mapsto f(x) \in f(X)$ is a homeomorphism.
- Let $f, g : X \to Y$ be two continuous maps. A *homotopy between f and g* is a continuous map $H : [0,1]\times X \to Y$ such that $H(0,\cdot) = f$ and $H(1,\cdot) = g$. If such a map exists, then f and g are said to be *homotopic*. Furthermore, if this map is proper, then f and g are said to be *properly homotopic*.
- Let $e^1, e^2 : X \to Y$ be two topological embeddings. An *ambient isotopy between e^1 and e^2* is a continuous map $\theta : [0,1] \times Y \to Y$ such that $\theta(0,\cdot) = \mathrm{id}_Y$, $\theta(1,\cdot)\circ e^1 = e^2$ and for every $t \in [0,1]$, $\theta(t,\cdot)$ is a homeomorphism. If such a map exists, then e^1 and e^2 are said to be *ambient isotopic within Y*.

Definition A.2. A topological space X is *contractible* if there exists $x^* \in X$ and a homotopy between the map $x \in X \mapsto x^* \in X$ and the identity map id_X on X.

Example A.1. 1. $\mathbb{R}^n$ and the unit n-hypercube $[0,1]^n$ are contractible. 2. The unit n-sphere $\mathbb{S}^n$ is not contractible.

Definition A.3. A topological space X is *paracompact* if every open cover of X admits a locally finite open subcover.

Example A.2. 1. Any compact space is paracompact. 2. Any subspace of $\mathbb{R}^n$ is paracompact (since it is metrizable; cf. Stone's theorem).

A.1.2. Fiber bundles, covering spaces and vector bundles

Definition A.4. Let E, F be a topological spaces, B be a connected topological space and $p : E \to B$ be a continuous map. The tuple (E, B, p, F) is called a *fiber bundle* (with *total space* E, *base space* B, *projection* p and *fiber* F) if for every $b \in B$, there exists an open subset V of B which contains b and a homeomorphism ψ making the following diagram commute:

$$\begin{tikzcd} p^{-1}(V) \arrow[r, "\psi"] \arrow[d, "p"] & V \times F \arrow[dl, "\mathrm{pr}_V"] \\ V & \end{tikzcd}$$

where $\mathrm{pr}_V : (x,y) \in V \times F \mapsto x \in V$. A *section of* (E, B, p, F) is a continuous map $s : B \to E$ such that $p\circ s = \mathrm{id}_B$.

Example A.3 (*Trivial Bundle*)**.** The *trivial bundle of F over B* is the fiber bundle $(B \times F, B, \mathrm{pr}_B, F)$.

Example A.4 (*Covering Space*)**.** If F is a discrete topological space, then (E, B, p, F) is called a *covering space*.

Example A.5 (*Vector Bundle*)**.** If $F = \mathbb{R}^n$, and the following conditions hold:

1. For every $b \in B$, $p^{-1}(b)$ is a n-dimensional (real) vector space.

2. For every $b \in B$, there exists an open subset V of B which containing b and a homeomorphism ψ making the diagram in Definition A.4 commute, such that for every $b' \in V$, the restricted map $x \in p^{-1}(b') \mapsto \psi(x) \in \{b'\} \times F$ is a linear isomorphism.

Then, (E, B, p, F) is called a *(real) vector bundle of rank n* and is denoted by (E, B, p, n).

Example A.6 (*Tangent Bundle*)**.** Let M be a smooth n-manifold (with or without boundary, with or without corners). The *tangent bundle of M* is the vector bundle $(\mathrm{T}M, M, p, n)$, where $\mathrm{T}M$ is the disjoint union of the tangent spaces $(\mathrm{T}_a M)_{a \in M}$ to M, and $p : (a, v) \in \mathrm{T}M \mapsto a \in M$ (for more details, see Appendix A.3).

Definition A.5. Let $(E, B, p, n), (E', B, p', n')$ be two vector bundles over the same base space. A continuous map $\zeta : E \to E'$ is a *vector bundle homomorphism* if the two following conditions hold:

1. The diagram:
$$\begin{tikzcd} E \arrow[r, "\zeta"] \arrow[d, "p"] & E' \arrow[dl, "p'"] \\ B & \end{tikzcd}$$
commutes.
2. For every $b \in B$, the restricted map $x \in p^{-1}(b) \mapsto \zeta(x) \in (p')^{-1}(b)$ is a linear map.

Moreover, the map ζ is a *vector bundle isomorphism* if ζ is a bijection, and both ζ and ζ^{-1} are vector bundle homomorphisms. In this case, (E, B, p, n) and (E', B, p', n') are said to be *isomorphic*.

Definition A.6. A vector bundle (E, B, p, n) is called *trivial* if it is isomorphic to the trivial vector bundle $(B \times \mathbb{R}^n, B, \mathrm{pr}_B, n)$ over B.

The following result can be found, for instance, in Bott and Tu (1982) — see Corollary 6.9, p. 59.

Theorem A.1. *Any vector bundle over a contractible paracompact base space is trivial.*

A.1.3. Topological degree of a proper continuous map between topological oriented n-manifolds

Let X be an oriented topological n-manifold and Y be an oriented connected topological n-manifold. For every proper continuous map $f : X \to Y$, one can associate an integer $\deg(f) \in \mathbb{Z}$ called the degree of f (see, e.g., Dold, 1995, Proposition and Definition 4.5, p. 268).

Proposition A.1. *Let $f, g : X \to Y$ be two proper continuous maps. If f and g are homotopic, then* $\deg(f) = \deg(g)$.

A.2. Differential calculus

Definition A.7. A function f from an arbitrary subset A of $\mathbb{R}^n$ to $\mathbb{R}$ is said to be C^1 if, for every $a \in A$, there exists an open subset V of $\mathbb{R}^n$ containing a, and a C^1 function $g : V \to \mathbb{R}$ such that $g|_{V \cap A} = f|_{V \cap A}$. Such a function is called a C^1 *extension of f around a*.

Proposition A.2. *Let C be a closed convex subset of $\mathbb{R}^n$ such that* $\mathrm{int}(C) \neq \emptyset$, *$f : C \to \mathbb{R}$ be a C^1 function, $c \in C$ and $g_1 : V_1 \to \mathbb{R}$, $g_2 : V_2 \to \mathbb{R}$ be two C^1 extensions of f around c. Then, for every $i = 1, \dots, n$,*
$$\frac{\partial g_1}{\partial x_i}(c) = \frac{\partial g_2}{\partial x_i}(c).$$

Definition A.8. Let C be a closed convex subset of $\mathbb{R}^n$ such that $\mathrm{int}(C) \neq \emptyset$, $f : C \to \mathbb{R}$ be a C^1 function, $c \in C$ and $g : V \to \mathbb{R}$ be a C^1 extension of f around c. For every $i = 1, \dots, n$,
$$\frac{\partial f}{\partial x_i}(c) = \frac{\partial g}{\partial x_i}(c).$$

A.3. Differential geometry

In the following, the term "smooth n-manifold" (without further qualification) refers to a smooth n-manifold with or without boundary, and with or without corners (see, e.g., Lee, 2012, pp. 10–15, p.25 and p. 415 for reminders about these definitions).

A.3.1. Tangent spaces and vector fields

Definition A.9. Let M be a smooth n-manifold and $a \in M$. A linear map $d : C^\infty(M, \mathbb{R}) \to \mathbb{R}$ is called a *derivation at a* if it satisfies the Leibniz rule: for every $f, g \in C^\infty(M, \mathbb{R})$, we have $d(fg) = d(f)g(a) + f(a)d(g)$. The vector space of all derivations at a is denoted by $\mathrm{T}_a M$ and is called the *tangent space to M at a*. Furthermore, the tuple $(\mathrm{T}M, M, p, n)$, where $\mathrm{T}M = \coprod_{a \in M} \mathrm{T}_a M$ and $p : (a, v) \in \mathrm{T}M \mapsto a \in M$, is a vector bundle of rank n which is called the *tangent bundle of M*.[11]

Definition A.10. Let M be a smooth n-manifold. A section $X : M \to \mathrm{T}M$ of the tangent bundle of M is called a *vector field on M*. The vector space of all vector fields on M is denoted $\mathfrak{X}(M)$. Moreover, for every vector field $X \in \mathfrak{X}(M)$ on M, the set $\mathfrak{Z}(X) = \{a \in M : X(a) = (a, \overrightarrow{0_a})\}$, where $\overrightarrow{0_a}$ denotes the derivation at a constantly equal to 0, is called the *set of zeros of X*. A zero $z \in \mathfrak{Z}(X)$ of X is called *isolated* if there exists an open neighborhood of z in M that does not intersect the set $\mathfrak{Z}(X) \backslash \{z\}$. A zero $z \in \mathfrak{Z}(X)$ of X is called *(manifold) interior* if there exists a chart (U, ϕ) of M such that $\phi(U)$ is an open subset of $\mathbb{R}^n$ (such a chart is called an *interior chart of M*) and $z \in U$.

A.3.2. Index of a zero of a vector field on a smooth manifold

Definition A.11. Let V be an open subset of $\mathbb{R}^n$ and $f : V \to \mathbb{R}^n$ be a continuous map. Moreover, let $z \in f^{-1}(0_{\mathbb{R}^n})$ be an isolated zero of f, i.e., there exists $\epsilon > 0$ such that $\overline{B}(z, \epsilon) \cap (f^{-1}(0_{\mathbb{R}^n}) \backslash \{z\}) = \emptyset$, where $\overline{B}(z, \epsilon) = \{y \in \mathbb{R}^n : \|z - y\|_2 \leq \epsilon\}$. The *index of f at z* is defined as the topological degree of the map
$$\begin{cases} f^z : & \partial\overline{B}(z, \epsilon) & \to & \mathbb{S}^{n-1} \\ & x & \mapsto & \frac{f(x)}{\|f(x)\|_2} \end{cases}$$
and is denoted by $\mathrm{ind}_z(f)$.[12]

Remark A.1 (*Canonical Vector Bundle Isomorphism*)**.** In the framework of Corollary 6.1 (see the beginning of Section 6), a particular vector bundle isomorphism from $\mathrm{T}M$ to $M \times \mathbb{R}^n$, denoted here $\zeta_{\mathfrak{c}}$ ($\mathfrak{c}$ standing for "canonical"), is usually considered when M is an open subset of $\mathbb{R}^n$. In this case, one can define
$$\begin{cases} \zeta_{\mathfrak{c}} : & \mathrm{T}M & \to & M \times \mathbb{R}^n \\ & (a, \frac{\partial}{\partial x_1}\Big|_a) & \mapsto & (a, e_1) = (a, (1, 0, 0, \dots)) \\ & (a, \frac{\partial}{\partial x_2}\Big|_a) & \mapsto & (a, e_2) = (a, (0, 1, 0, \dots)) \\ & \dots & & \\ & (a, \frac{\partial}{\partial x_n}\Big|_a) & \mapsto & (a, e_n) = (a, (\dots, 0, 0, 1)) \end{cases},$$
where for every $a \in M$, $\{\frac{\partial}{\partial x_1}\Big|_a, \dots, \frac{\partial}{\partial x_n}\Big|_a\}$ is the canonical basis of $\mathrm{T}_a M$, with for every $i \in \{1, \dots, n\}$, $\frac{\partial}{\partial x_i}\Big|_a : f \in C^\infty(M, \mathbb{R}) \mapsto \frac{\partial f}{\partial x_i}(a) \in \mathbb{R}$. In that case, the bijection $\Psi_{\zeta_{\mathfrak{c}}} : C^0(M, \mathbb{R}^n) \to \mathfrak{X}(M)$ defined in the proof of Corollary 6.1 is simply denoted $\Psi_{\mathfrak{c}}$.

[11] See, e.g., Lee (2012), Proposition 3.18, p. 66 and Proposition 10.4, p. 252 for reminders about the natural topology on $\mathrm{T}M$.

[12] See Appendix A.1 for reminders about topological degree of a proper continuous map. Note that the (continuous) map f^z is proper since it maps the compact space $\partial\overline{B}(z, \epsilon)$ to the Hausdorff space $\mathbb{S}^{n-1}$. Moreover, the index $\mathrm{ind}_z(f)$ of f at z does not depend on ϵ (see, e.g., Guillemin and Pollack, 1975, p. 133).

Definition A.12. Let V be an open subset of $\mathbb{R}^n$, $X : V \to \mathrm{T}V$ be a vector field on V and $z \in \mathfrak{Z}(X)$ be an isolated zero of X. The *index of X at z* is defined as the index of $\Psi_{\mathfrak{c}}^{-1}(X)$ at z and is denoted by $\mathrm{ind}_z(X)$, i.e., $\mathrm{ind}_z(X) = \mathrm{ind}_z(\Psi_{\mathfrak{c}}^{-1}(X))$.[13]

Definition A.13. Let M, N be two smooth manifolds, $X \in \mathfrak{X}(M)$ be a vector field on M, $X' \in \mathfrak{X}(N)$ be a vector field on N, and $f : M \to N$ be a smooth map. The maps X and X' are said to be *f-related* if the following diagram commutes:

$$\begin{array}{ccc} \mathrm{T}M & \xrightarrow{\mathrm{T}f} & \mathrm{T}N \\ {\scriptstyle X}\uparrow & & {\scriptstyle X'}\uparrow \\ M & \xrightarrow{f} & N \end{array}$$

where $\mathrm{T}f : (a,d) \in \mathrm{T}M \mapsto (f(a), \mathrm{T}_a f(d)) \in \mathrm{T}N$ is the global tangent map of f, with $\mathrm{T}_a f(d)(g) = d(g \circ f) \in \mathbb{R}$, for every $g \in C^\infty(N, \mathbb{R})$ ($\mathrm{T}_a f(d) \in \mathrm{T}_{f(a)} N$).

Definition A.14. Let M, N be two smooth manifolds, $X \in \mathfrak{X}(M)$ be a vector field on M and $f : M \to N$ be a diffeomorphism. There exists a unique vector field $f_* X \in \mathfrak{X}(N)$ on N, called the *pushforward of X by f*, such that X and $f_* X$ are f-related, which is defined as follows: for every $a \in N$, $f_* X(a) = (\mathrm{T}_{f^{-1}(a)} f \circ X \circ f^{-1})(a)$.

Definition A.15. Let M be a smooth n-manifold, $X \in \mathfrak{X}(M)$ be a vector field on M, $z \in \mathfrak{Z}(X)$ be an interior isolated zero of X and (U, ϕ) be an interior chart of M such that $z \in U$. The *index of X at z* is defined as the index of the pushforward $\phi_* X$ of X by ϕ at $\phi(z)$, i.e., the index of the vector field

$$\left\{ \begin{array}{cccc} \phi_* X : & \phi(U) & \to & \mathrm{T}\phi(U) \\ & a & \mapsto & (\mathrm{T}_{\phi^{-1}(a)} \phi \circ X \circ \phi^{-1})(a) \end{array} \right.$$

(on the open subset $\phi(U)$ of $\mathbb{R}^n$) at $\phi(z)$, and is denoted by $\mathrm{ind}_z(X)$.[14]

A.4. Real semialgebraic geometry

Definition A.16. A subset S of $\mathbb{R}^n$ is said to be *semialgebraic* if there exists some integer $p \geq 1$, integers $r_i \geq 1$ for each $1 \leq i \leq p$, and polynomial $P_{i,j} \in \mathbb{R}[X_1, \ldots, X_n]$ for each $1 \leq i \leq p$ and each $1 \leq j \leq r_i$, such that $S = \bigcup_{i=1}^p \bigcap_{j=1}^{r_i} \{x \in \mathbb{R}^n : P_{i,j}(x) \geq 0\}$.

Example A.7. 1. The unit n-disk $\mathbb{D}^n = \{(x_1, \ldots, x_{n+1}) \in \mathbb{R}^{n+1} : \sum_{i=1}^{n+1} x_i^2 \leq 1\}$, the unit n-sphere $\mathbb{S}^n = \{(x_1, \ldots, x_{n+1}) \in \mathbb{R}^{n+1} : \sum_{i=1}^{n+1} x_i^2 = 1\}$ and the unit n-simplex $\Delta^n = \{(x_1, \ldots, x_{n+1}) \in \mathbb{R}^{n+1} : \sum_{k=i}^{n+1} x_i = 1 \text{ and } \forall i \in \{1, \ldots, n+1\}, x_i \geq 0\}$ are semialgebraic. 2. The graph $\{(x, y) \in \mathbb{R}^2 : y = \cos(x)\}$ of the cosine function is not semialgebraic.

Definition A.17. Let S be a semialgebraic subset of $\mathbb{R}^n$ and T be a semialgebraic subset of $\mathbb{R}^m$. A map $f : S \to T$ is said to be *semialgebraic* if its graph $\mathrm{Gr}(f)$ is a semialgebraic subset of $\mathbb{R}^n \times \mathbb{R}^m$.

Example A.8. 1. If $S \subset \mathbb{R}^n$ and $T \subset \mathbb{R}^m$ are semialgebraic, and if $f : S \to T$ is a polynomial function, then f is semialgebraic. 2. The map $f : x \in \mathbb{R} \backslash \{0\} \mapsto \frac{1}{x} \in \mathbb{R}$ is semialgebraic since

$$\mathrm{Gr}(f) = \{(x, y) \in \mathbb{R} \backslash \{0\} \times \mathbb{R} : y = \frac{1}{x}\} = \{(x, y) \in \mathbb{R}^2 : x \neq 0 \text{ and } xy = 1\}.$$

For every semialgebraic set S, one can associate an integer $\dim(S) \in \mathbb{N}$ called the dimension of S (see, e.g., Bochnak et al., 1998, Definition 2.8.1, p. 50).

[13] Here, V is viewed as an embedded n-submanifold of $\mathbb{R}^n$ (see, e.g., Lee, 2012, Proposition 5.1, p. 99). Also, the tangent bundle of V is trivial (see, e.g., Lee, 2012, Proposition 3.20, p. 67).

[14] The quantity $\mathrm{ind}_z(X)$ does not depend on the chart (U, ϕ) (see, e.g., Guillemin and Pollack, 1975, p. 134).

Table B.2
Table of notations for the unknottedness theorem.

$\mathbf{P}_{\mathcal{V}}$	*Graph of pairwise stable networks associated with a set $\mathcal{V}$ of societies*
$\pi_{\mathbf{G}} : \mathbf{P}_{\mathcal{V}} \to \mathbf{G}$	*Projection from $\mathbf{P}_{\mathcal{V}} \subset \mathcal{V} \times \mathbf{G}$ to $\mathbf{G}$, where $\mathcal{V}$ is a set of societies*
$\eta_{\mathcal{V}} : \mathbf{P}_{\mathcal{V}} \to \mathcal{V}$	*Homeomorphism (with inverse $\rho_{\mathcal{V}} : \mathcal{V} \to \mathbf{P}_{\mathcal{V}}$) of Bich-Fixary's topological structure theorem, where $\mathcal{V}$ is a regular set of societies*
$\widetilde{\mathbf{E}}_{\mathcal{V}}$	*ε-ambient space $\mathcal{V} \times \widetilde{\mathbf{G}}$ associated with a set $\mathcal{V}$ of societies*
$\kappa_{\mathcal{V}}^0 : \mathcal{V} \to \widetilde{\mathbf{E}}_{\mathcal{V}}$	*Unknot associated with $\mathcal{V}$ (and g^0), where $\mathcal{V}$ is a set of societies*
$\kappa_{\mathcal{V}} : \mathcal{V} \to \widetilde{\mathbf{E}}_{\mathcal{V}}$	*Knot of pairwise stable networks associated with $\mathcal{V}$ (and $\rho_{\mathcal{V}}$), where $\mathcal{V}$ is a regular set of societies*

Definition A.18. Let S be a semialgebraic set. A semialgebraic set $G \subset S$ is said to be *generic in S* (or a *generic subset of S*) if:

1. $\dim(S \backslash G) < \dim(S)$.
2. G is open in S.

Example A.9. The complement of a line in the plane $\mathbb{R}^2$ is a generic subset of $\mathbb{R}^2$.

Appendix B. Proof of the unknottedness theorem

We provide in Table B.2 a list of the notations used throughout the proof.

Recall that $\varepsilon > 0$ and $g^0 \in \mathbf{G}$ are fixed. Consider a fixed regular set $\mathcal{V}$ of societies.Recall also that the homeomorphism $\eta_{\mathcal{V}} : \mathbf{P}_{\mathcal{V}} \to \mathcal{V}$ in Bich–Fixary's topological structure theorem (see Theorem 4.1 in Section 4) is defined as follows:

$$\left\{ \begin{array}{cccc} \eta_{\mathcal{V}} : & \mathbf{P}_{\mathcal{V}} & \to & \mathcal{V} \\ & (v, g) & \mapsto & \overline{u}^{v,g} \end{array} \right. ,$$

where for every $i \in I$ and every $\gamma \in \mathbf{G}$,

$$\overline{u}_i^{v,g}(\gamma) = v_i(\gamma) + \sum_{j \neq i} \big(\partial_{ij} v_i(g_{ij}, g_{-ij}) - \partial_{ij} v_i(g_{ij}, g_{-ij}^0)\big)(\gamma_{ij} - g_{ij}) + \sum_{j \neq i} g_{ij} \gamma_{ij}.$$

The inverse of $\eta_{\mathcal{V}}$ is defined as

$$\left\{ \begin{array}{cccc} \rho_{\mathcal{V}} : & \mathcal{V} & \to & \mathbf{P}_{\mathcal{V}} \\ & v & \mapsto & (\underline{u}^v, g^v) \end{array} \right. ,$$

where for every $i \in I$ and every $\gamma \in \mathbf{G}$,

$$\underline{u}_i^v(\gamma) = v_i(\gamma) - \sum_{j \neq i} \big(\partial_{ij} v_i(g_{ij}^v, g_{-ij}^v) - \partial_{ij} v_i(g_{ij}^v, g_{-ij}^0)\big)(\gamma_{ij} - g_{ij}^v) - \sum_{j \neq i} g_{ij}^v \gamma_{ij},$$

and where for every link $ij \in \mathbf{L}$, $g_{ij}^v = \min\{w_{i,j}^v, w_{j,i}^v\}$, with $w_{i,j}^v \in [0,1]$ being the unique maximizer of the strictly concave function $u_i^v : w \in [0,1] \mapsto v_i(w, g_{-ij}^0) - \frac{w^2}{2} \in \mathbb{R}$ and $w_{j,i}^v \in [0,1]$ being the unique maximizer of the strictly concave function $u_j^v : w \in [0,1] \mapsto v_j(w, g_{-ij}^0) - \frac{w^2}{2} \in \mathbb{R}$.

For every agent $i \in I$ and every $(v, g) \in \mathcal{V} \times \mathbf{G}$, define the maps

$$\left\{ \begin{array}{cccc} h_i^{v,g} : & \mathbf{G} & \to & \mathbb{R} \\ & \gamma & \mapsto & \sum_{j \neq i} \big(\partial_{ij} v_i(g_{ij}, g_{-ij}) - \partial_{ij} v_i(g_{ij}, g_{-ij}^0)\big)(\gamma_{ij} - g_{ij}) \end{array} \right.$$

and

$$\left\{ \begin{array}{cccc} l_i^g : & \mathbf{G} & \to & \mathbb{R} \\ & \gamma & \mapsto & \sum_{j \neq i} \gamma_{ij} g_{ij} \end{array} \right. ,$$

so that $\overline{u}_i^{v,g} = v_i + h_i^{v,g} + l_i^g$ and $\underline{u}_i^v = v_i - h_i^{v,g^v} - l_i^{g^v}$.

Step I. Construction of an ambient isotopy between $\kappa_{\mathcal{V}} : \mathcal{V} \to \widetilde{\mathbf{E}}_{\mathcal{V}}$ and $e^{\pi_{\mathbf{G}} \circ \rho_{\mathcal{V}}} : v \in \mathcal{V} \mapsto (v, (\pi_{\mathbf{G}} \circ \rho_{\mathcal{V}})(v)) \in \widetilde{\mathbf{E}}_{\mathcal{V}}$. We show that the knot $\kappa_{\mathcal{V}} : \mathcal{V} \to \widetilde{\mathbf{E}}_{\mathcal{V}}$ of pairwise stable networks associated with $\mathcal{V}$ is ambient isotopic to the topological embedding $e^{\pi_{\mathbf{G}} \circ \rho_{\mathcal{V}}} : v \in \mathcal{V} \mapsto \big(v, (\pi_{\mathbf{G}} \circ \rho_{\mathcal{V}})(v)\big) \in \widetilde{\mathbf{E}}_{\mathcal{V}}$ within $\widetilde{\mathbf{E}}_{\mathcal{V}}$, through an ambient isotopy $\overline{\theta}_{\mathcal{V}}^1$. Let r be the projection from $\widetilde{\mathbf{G}}$ to the closest point in $\mathbf{G}$ (this map is a retraction of $\widetilde{\mathbf{G}}$ on $\mathbf{G}$, i.e., a continuous map such that its composition with the

inclusion $\mathbf{G} \hookrightarrow \widetilde{\mathbf{G}}$ is the identity map on $\mathbf{G}$), and $b : \widetilde{\mathbf{G}} \to [0,1]$ be a continuous map such that $b(\mathbf{G}) = 1$ and $b(\partial\widetilde{\mathbf{G}}) = 0$. Now, let

$$\left\{ \begin{array}{lccc} \overline{\theta}^1_{\mathcal{V}} : & [0,1]\times\widetilde{\mathbf{E}}_{\mathcal{V}} & \to & \widetilde{\mathbf{E}}_{\mathcal{V}} \\ & (t,(v,g)) & \mapsto & (\overline{u}^{t,v,g}, g) \end{array} \right. ,$$

where for every $i \in I$, $\overline{u}_i^{t,v,g} = v_i + tb(g)(h_i^{v,r(g)} + l_i^{r(g)})$, and let

$$\left\{ \begin{array}{lccc} \underline{\theta}^1_{\mathcal{V}} : & [0,1]\times\widetilde{\mathbf{E}}_{\mathcal{V}} & \to & \widetilde{\mathbf{E}}_{\mathcal{V}} \\ & (t,(v,g)) & \mapsto & (\underline{u}^{t,v,g}, g) \end{array} \right. ,$$

where for every $i \in I$, $\underline{u}_i^{t,v,g} = v_i - tb(g)(h_i^{v,r(g)} + l_i^{r(g)})$. Note that due to $\mathcal{A}$-invariance assumption, both $\overline{\theta}^1_{\mathcal{V}}$ and $\underline{\theta}^1_{\mathcal{V}}$ are well-defined maps.

First, $\overline{\theta}^1_{\mathcal{V}}$ is a continuous map since both r and b are continuous maps (for more details, see the proof of the continuity of $\eta_{\mathcal{V}}$ in Bich and Fixary, 2022, whose proof follows a similar argument).

Second, for every $i \in I$ and every $(v,g) \in \widetilde{\mathbf{E}}_{\mathcal{V}}$,

$$\overline{u}_i^{0,v,g} = v_i + 0b(g)(h_i^{v,r(g)} + l_i^{r(g)}) = v_i.$$

Thus, $\overline{\theta}^1_{\mathcal{V}}(0,\cdot,\cdot) = \mathrm{id}_{\widetilde{\mathbf{E}}_{\mathcal{V}}}$.

Third, for every $v \in \mathcal{V}$,

$$(\overline{\theta}^1_{\mathcal{V}}(1,\cdot,\cdot)\circ\kappa_{\mathcal{V}})(v) = \overline{\theta}^1_{\mathcal{V}}(1,\rho_{\mathcal{V}}(v)) = \overline{\theta}^1_{\mathcal{V}}(1,(\underline{u}^v, g^v)) = (\overline{u}^{1,\underline{u}^v,g^v}, g^v),$$

where for every $i \in I$,

$$\begin{aligned} \overline{u}_i^{1,\underline{u}^v,g^v} &= \underline{u}_i^v + 1b(g^v)(h_i^{\underline{u}^v,r(g^v)} + l_i^{r(g^v)}) \\ &= (v_i - h_i^{v,g^v} - l_i^{g^v}) + b(g^v)(h_i^{\underline{u}^v,r(g^v)} + l_i^{r(g^v)}) \\ &= (v_i - h_i^{v,g^v} - l_i^{g^v}) + (h_i^{\underline{u}^v,r(g^v)} + l_i^{r(g^v)}) \text{ (since } g^v \in \mathbf{G},\ b(g^v) = 1) \\ &= (v_i - h_i^{v,g^v} - l_i^{g^v}) + (h_i^{\underline{u}^v,g^v} + l_i^{g^v}) \text{ (since } g^v \in \mathbf{G},\ r(g^v) = g^v) \\ &= v_i - h_i^{v,g^v} + h_i^{\underline{u}^v,g^v}. \end{aligned}$$

Observe that for every $\gamma \in \mathbf{G}$,

$$\begin{aligned} h_i^{\underline{u}^v,g^v}(\gamma) &= \sum_{j\neq i}\big(\partial_{ij}\underline{u}_i^v(g_{ij}^v, g_{-ij}^v) - \partial_{ij}\underline{u}_i^v(g_{ij}^v, g_{-ij}^0)\big)(\gamma_{ij} - g_{ij}^v) \\ &= \sum_{j\neq i}\big(\partial_{ij}(v_i - h_i^{v,g^v} - l_i^{g^v})(g_{ij}^v, g_{-ij}^v) \\ &\quad - \partial_{ij}(v_i - h_i^{v,g^v} - l_i^{g^v})(g_{ij}^v, g_{-ij}^0)\big)(\gamma_{ij} - g_{ij}^v) \\ &= \sum_{j\neq i}\big(\partial_{ij}v_i(g_{ij}^v, g_{-ij}^v) - \partial_{ij}v_i(g_{ij}^v, g_{-ij}^0)\big)(\gamma_{ij} - g_{ij}^v) \\ &= h_i^{v,g^v}(\gamma), \end{aligned}$$

since both $\partial_{ij}h_i^{v,g^v}$ and $\partial_{ij}l_i^{g^v}$ are constant maps. Thus, $\overline{\theta}^1_{\mathcal{V}}(1,\cdot,\cdot)\circ\kappa_{\mathcal{V}} = e^{\pi_{\mathbf{G}}\circ\rho_{\mathcal{V}}}$.

Finally, we show that for every $t \in [0,1]$, $\overline{\theta}^1_{\mathcal{V}}(t,\cdot,\cdot)$ is a homeomorphism with inverse $\underline{\theta}^1_{\mathcal{V}}(t,\cdot,\cdot)$. Let $t \in [0,1]$.

First, both $\overline{\theta}^1_{\mathcal{V}}(t,\cdot,\cdot)$ and $\underline{\theta}^1_{\mathcal{V}}(t,\cdot,\cdot)$ are continuous maps (since both $\overline{\theta}^1_{\mathcal{V}}$ and $\underline{\theta}^1_{\mathcal{V}}$ are continuous by construction).

Second, for every $(v,g) \in \widetilde{\mathbf{E}}_{\mathcal{V}}$,

$$(\overline{\theta}^1_{\mathcal{V}}(t,\cdot,\cdot)\circ\underline{\theta}^1_{\mathcal{V}}(t,\cdot,\cdot))(v,g) = \overline{\theta}^1_{\mathcal{V}}(t,(\underline{u}^{t,v,g}, g)) = (\overline{u}^{t,\underline{u}^{t,v,g},g}, g),$$

where for every $i \in I$,

$$\begin{aligned} \overline{u}_i^{t,\underline{u}^{t,v,g},g} &= \underline{u}_i^{t,v,g} + tb(g)(h_i^{\underline{u}^{t,v,g},r(g)} + l_i^{r(g)}) \\ &= v_i - tb(g)(h_i^{v,r(g)} + l_i^{r(g)}) + tb(g)(h_i^{\underline{u}^{t,v,g},r(g)} + l_i^{r(g)}) \\ &= v_i + tb(g)(h_i^{\underline{u}^{t,v,g},r(g)} - h_i^{v,r(g)}). \end{aligned}$$

Observe that for every $\gamma \in \mathbf{G}$,

$$\begin{aligned} & h_i^{\underline{u}^{t,v,g},r(g)}[\underline{u}^{t,v,g}](\gamma) \\ &= \sum_{j\neq i}\big(\partial_{ij}\underline{u}_i^{t,v,g}(r(g)_{ij}, r(g)_{-ij}) - \partial_{ij}\underline{u}_i^{t,v,g}(r(g)_{ij}, g_{-ij}^0)\big)(\gamma_{ij} - r(g)_{ij}) \\ &= \sum_{j\neq i}\big(\ \partial_{ij}(v_i - tb(g)(h_i^{v,r(g)} + l_i^{r(g)}))(r(g)_{ij}, r(g)_{-ij}) \\ &\quad - \partial_{ij}(v_i - tb(g)(h_i^{v,r(g)} + l_i^{r(g)}))(r(g)_{ij}, g_{-ij}^0)\ \big)(\gamma_{ij} - r(g)_{ij}) \\ &= \sum_{j\neq i}\big(\partial_{ij}v_i(r(g)_{ij}, r(g)_{-ij}) - \partial_{ij}v_i(r(g)_{ij}, g_{-ij}^0)\big)(\gamma_{ij} - r(g)_{ij}) \\ &= h_i^{v,r(g)}(\gamma), \end{aligned}$$

since both $\partial_{ij}h_i^{v,r(g)}$ and $\partial_{ij}l_i^{r(g)}$ are constant maps. This implies that $\overline{u}^{t,\underline{u}^{t,v,g},g} = v$.

Similarly, one can verify that for every $(v,g) \in \widetilde{\mathbf{E}}_{\mathcal{V}}$,

$$(\underline{\theta}^1_{\mathcal{V}}(t,\cdot,\cdot)\circ\overline{\theta}^1_{\mathcal{V}}(t,\cdot,\cdot))(v,g) = \underline{\theta}^1_{\mathcal{V}}(t,(\overline{u}^{t,v,g}, g)) = (\underline{u}^{t,\overline{u}^{t,v,g},g}, g) = (v,g).$$

Finally, from the previous points, $\overline{\theta}^1_{\mathcal{V}}$ is, by definition, an ambient isotopy between $\kappa_{\mathcal{V}}$ and the topological embedding $e^{\pi_{\mathbf{G}}\circ\rho_{\mathcal{V}}}$ within the ambient space $\widetilde{\mathbf{E}}_{\mathcal{V}}$. Note that for every $t \in [0,1]$, $\overline{\theta}^1_{\mathcal{V}}(t,\cdot,\cdot)|_{\mathcal{V}\times\partial\widetilde{\mathbf{G}}} = \mathrm{id}_{\mathcal{V}\times\partial\widetilde{\mathbf{G}}}$, i.e., the ambient isotopy $\overline{\theta}^1_{\mathcal{V}}$ does not deform the boundary of the ambient space.

Step II. Construction of an ambient isotopy between $e^f : v \in \mathcal{V} \mapsto (v, f^v) \in \widetilde{\mathbf{E}}_{\mathcal{V}}$ and $\kappa^0_{\mathcal{V}} : \mathcal{V} \to \widetilde{\mathbf{E}}_{\mathcal{V}}$. We show that for every continuous map $f : v \in \mathcal{V} \mapsto f^v \in \mathbf{G}$, the topological embedding $e^f : v \in \mathcal{V} \mapsto (v, f^v) \in \widetilde{\mathbf{E}}_{\mathcal{V}}$ is ambient isotopic to the unknot $\kappa^0_{\mathcal{V}} : \mathcal{V} \to \widetilde{\mathbf{E}}_{\mathcal{V}}$ associated with $\mathcal{V}$ within $\widetilde{\mathbf{E}}_{\mathcal{V}}$, through an ambient isotopy $\overline{\theta}^{2,f}_{\mathcal{V}}$. For every link $ij \in \mathbf{L}$ and every $w \in [0,1]$, consider the piecewise linear maps $\overline{\chi}^w_{ij}$ and $\underline{\chi}^w_{ij}$ defined by

$$\left\{ \begin{array}{lccc} \overline{\chi}^w_{ij} : & [-\varepsilon, 1+\varepsilon] & \to & [-\varepsilon, 1+\varepsilon] \\ & -\varepsilon & \mapsto & -\varepsilon \\ & w & \mapsto & g_{ij}^0 \\ & 1+\varepsilon & \mapsto & 1+\varepsilon \end{array} \right.$$

and

$$\left\{ \begin{array}{lccc} \underline{\chi}^w_{ij} : & [-\varepsilon, 1+\varepsilon] & \to & [-\varepsilon, 1+\varepsilon] \\ & -\varepsilon & \mapsto & -\varepsilon \\ & g_{ij}^0 & \mapsto & w \\ & 1+\varepsilon & \mapsto & 1+\varepsilon \end{array} \right. .$$

Furthermore, for every $ij \in \mathbf{L}$, consider the maps

$$\overline{\chi}_{ij} : (w, \widetilde{w}) \in [0,1]\times[-\varepsilon, 1+\varepsilon] \mapsto \overline{\chi}^w_{ij}(\widetilde{w}) \in [-\varepsilon, 1+\varepsilon]$$

and

$$\underline{\chi}_{ij} : (w, \widetilde{w}) \in [0,1]\times[-\varepsilon, 1+\varepsilon] \mapsto \underline{\chi}^w_{ij}(\widetilde{w}) \in [-\varepsilon, 1+\varepsilon],$$

and observe that:

1. **Property 1:** For every $w \in [0,1]$, $\overline{\chi}^w_{ij}$ is a homeomorphism with inverse $\underline{\chi}^w_{ij}$.
2. **Property 2:** $\overline{\chi}^{g_{ij}^0}_{ij} = \underline{\chi}^{g_{ij}^0}_{ij} = \mathrm{id}_{[-\varepsilon,1+\varepsilon]}$.

Now, let

$$\left\{ \begin{array}{lccc} \overline{\theta}^{2,f}_{\mathcal{V}} : & [0,1]\times\widetilde{\mathbf{E}}_{\mathcal{V}} & \to & \widetilde{\mathbf{E}}_{\mathcal{V}} \\ & (t,(v,g)) & \mapsto & (v, \overline{\gamma}^{t,v,g}) \end{array} \right. ,$$

where for every $ij \in \mathbf{L}$, $\overline{\gamma}^{t,v,g}_{ij} = \overline{\chi}_{ij}((1-t)g_{ij}^0 + tf_{ij}^v, g_{ij})$, and let

$$\left\{ \begin{array}{lccc} \underline{\theta}^{2,f}_{\mathcal{V}} : & [0,1]\times\widetilde{\mathbf{E}}_{\mathcal{V}} & \to & \widetilde{\mathbf{E}}_{\mathcal{V}} \\ & (t,(v,g)) & \mapsto & (v, \underline{\gamma}^{t,v,g}) \end{array} \right. ,$$

where for every $ij \in \mathbf{L}$, $\underline{\gamma}^{t,v,g}_{ij} = \underline{\chi}_{ij}((1-t)g_{ij}^0 + tf_{ij}^v, g_{ij})$.

First, $\overline{\theta}^{2,f}_{\mathcal{V}}$ is continuous as the composition of continuous maps (recall that by assumption, the map $f : \mathcal{V} \to \mathbf{G}$ is continuous).

Second, for every $ij \in \mathbf{L}$ and every $(v,g) \in \widetilde{\mathbf{E}}_{\mathcal{V}}$,

$$\overline{\gamma}^{0,v,g}_{ij} = \overline{\chi}_{ij}((1-0)g_{ij}^0 + 0f_{ij}^v, g_{ij}) = \overline{\chi}_{ij}(g_{ij}^0, g_{ij}).$$

From Property 2, $\overline{\chi}_{ij}^{g_{ij}^0} = \overline{\chi}_{ij}(g_{ij}^0, \cdot)$ is the identity map on $[-\varepsilon, 1+\varepsilon]$. Hence, $\overline{\theta}_{\mathcal{V}}^{2,f}(0,\cdot,\cdot) = \mathrm{id}_{\widetilde{\mathbf{E}}_{\mathcal{V}}}$.

Third, for every $v \in \mathcal{V}$,

$$(\overline{\theta}_{\mathcal{V}}^{2,f}(1,\cdot,\cdot)\circ e^f)(v) = \overline{\theta}_{\mathcal{V}}^{2,f}(1,(v,f^v)) = (v, \overline{\gamma}^{1,v,f^v}),$$

where for every $ij \in \mathbf{L}$,

$$\begin{aligned}\overline{\gamma}_{ij}^{1,v,f^v} &= \overline{\chi}_{ij}((1-1)g_{ij}^0 + 1f_{ij}^v, f_{ij}^v)\\ &= \overline{\chi}_{ij}^{f_{ij}^v}(f_{ij}^v)\\ &= g_{ij}^0 \text{ (by definition of } \overline{\chi}_{ij})\end{aligned}$$

Thus, $\overline{\theta}_{\mathcal{V}}^{2,f}(1,\cdot,\cdot)\circ e^f = \kappa_{\mathcal{V}}^0$.

Finally, it remains to show that for every $t \in [0,1]$, $\overline{\theta}_{\mathcal{V}}^{2,f}(t,\cdot,\cdot)$ is a homeomorphism with inverse $\underline{\theta}_{\mathcal{V}}^{2,f}(t,\cdot,\cdot)$. Let $t \in [0,1]$.

First, both $\overline{\theta}_{\mathcal{V}}^{2,f}(t,\cdot,\cdot)$ and $\underline{\theta}_{\mathcal{V}}^{2,f}(t,\cdot,\cdot)$ are continuous maps (as both $\overline{\theta}_{\mathcal{V}}^{2,f}$ and $\underline{\theta}_{\mathcal{V}}^{2,f}$ are continuous maps).

Second, for every $(v,g) \in \widetilde{\mathbf{E}}_{\mathcal{V}}$,

$$(\overline{\theta}_{\mathcal{V}}^{2,f}(t,\cdot,\cdot)\circ\underline{\theta}_{\mathcal{V}}^{2,f}(t,\cdot,\cdot))(v,g) = \overline{\theta}_{\mathcal{V}}^{2,f}(t,(v,\underline{\gamma}^{t,v,g})) = (v, \overline{\gamma}^{t,v,\underline{\gamma}^{t,v,g}}),$$

where for every $ij \in \mathbf{L}$,

$$\begin{aligned}\overline{\gamma}_{ij}^{t,v,\underline{\gamma}^{t,v,g}} &= \overline{\chi}_{ij}((1-t)g_{ij}^0 + tf_{ij}^v, \underline{\gamma}_{ij}^{t,v,g})\\ &= \overline{\chi}_{ij}((1-t)g_{ij}^0 + tf_{ij}^v, \underline{\chi}_{ij}((1-t)g_{ij}^0 + tf_{ij}^v, g_{ij}))\\ &= (\overline{\chi}_{ij}^{(1-t)g_{ij}^0+tf_{ij}^v}\circ\underline{\chi}_{ij}^{(1-t)g_{ij}^0+tf_{ij}^v})(g_{ij})\\ &= g_{ij} \text{ (from Property 1).}\end{aligned}$$

Similarly, one can verify that for every $(v,g) \in \widetilde{\mathbf{E}}_{\mathcal{V}}$,

$$(\underline{\theta}_{\mathcal{V}}^{2,f}(t,\cdot,\cdot)\circ\overline{\theta}_{\mathcal{V}}^{2,f}(t,\cdot,\cdot))(v,g) = \underline{\theta}_{\mathcal{V}}^{2,f}(t,(v,\overline{\gamma}^{t,v,g})) = (v, \underline{\gamma}^{t,v,\overline{\gamma}^{t,v,g}}) = (v,g).$$

Finally, from the previous points, $\overline{\theta}_{\mathcal{V}}^{2,f}$ is, by definition, an ambient isotopy between e^f and the unknot $\kappa_{\mathcal{V}}^0$ within the ambient space $\widetilde{\mathbf{E}}_{\mathcal{V}}$. Remark that for every $t \in [0,1]$, $\overline{\theta}_{\mathcal{V}}^{2,f}(t,\cdot,\cdot)|_{\mathcal{V}\times\partial\widetilde{\mathbf{G}}} = \mathrm{id}_{\mathcal{V}\times\partial\widetilde{\mathbf{G}}}$, i.e., the ambient isotopy $\overline{\theta}_{\mathcal{V}}^{2,f}$ does not deform the boundary of the ambient space.

Step III. Construction of an ambient isotopy between $\kappa_{\mathcal{V}}$ and $\kappa_{\mathcal{V}}^0$ — The unknottedness theorem. To finish, from Step I and Step II, notice that the map

$$\begin{cases}\theta_{\mathcal{V}}: & [0,1]\times\widetilde{\mathbf{E}}_{\mathcal{V}} \to \widetilde{\mathbf{E}}_{\mathcal{V}}\\ & (t,(v,g)) \mapsto (\overline{\theta}_{\mathcal{V}}^{2,\pi_{\mathbf{G}}\circ\rho_{\mathcal{V}}}(t,\cdot,\cdot)\circ\overline{\theta}_{\mathcal{V}}^{1}(t,\cdot,\cdot))(v,g)\end{cases}$$

is an ambient isotopy between $\kappa_{\mathcal{V}}$ and $\kappa_{\mathcal{V}}^0$ within $\widetilde{\mathbf{E}}_{\mathcal{V}}$. Indeed:

1. $\theta_{\mathcal{V}}$ is continuous as the composition of continuous maps.
2. $\overline{\theta}_{\mathcal{V}}^{2,\pi_{\mathbf{G}}\circ\rho_{\mathcal{V}}}(0,\cdot,\cdot)\circ\overline{\theta}_{\mathcal{V}}^{1}(0,\cdot,\cdot) = \mathrm{id}_{\widetilde{\mathbf{E}}_{\mathcal{V}}}\circ\mathrm{id}_{\widetilde{\mathbf{E}}_{\mathcal{V}}} = \mathrm{id}_{\widetilde{\mathbf{E}}_{\mathcal{V}}}$.
3. $(\overline{\theta}_{\mathcal{V}}^{2,\pi_{\mathbf{G}}\circ\rho_{\mathcal{V}}}(1,\cdot,\cdot)\circ\overline{\theta}_{\mathcal{V}}^{1}(1,\cdot,\cdot))\circ\kappa_{\mathcal{V}} = \overline{\theta}_{\mathcal{V}}^{2,\pi_{\mathbf{G}}\circ\rho_{\mathcal{V}}}(1,\cdot,\cdot)\circ e^{\pi_{\mathbf{G}}\circ\rho_{\mathcal{V}}} = \kappa_{\mathcal{V}}^0$.
4. For every $t \in [0,1]$, $\theta_{\mathcal{V}}(t,\cdot,\cdot)$ is a homeomorphism with inverse $\underline{\theta}_{\mathcal{V}}^{1}(t,\cdot,\cdot)\circ\underline{\theta}_{\mathcal{V}}^{2,\pi_{\mathbf{G}}\circ\rho_{\mathcal{V}}}(t,\cdot,\cdot)$.

Additionally, from Step I and Step II, observe that for every $t \in [0,1]$, $\theta_{\mathcal{V}}(t,\cdot,\cdot)|_{\mathcal{V}\times\partial\widetilde{\mathbf{G}}} = \mathrm{id}_{\mathcal{V}\times\partial\widetilde{\mathbf{G}}}$, i.e., the ambient isotopy $\theta_{\mathcal{V}}$ does not deform the boundary of the ambient space. This step concludes the proof of the unknottedness theorem. □

Appendix C. Proofs of Proposition 6.1 and of Corollary 6.1

C.1. Proof of Proposition 6.1

By definition, since $\mathrm{T}M$ is trivial, there exists a vector bundle isomorphism from $(\mathrm{T}M, M, p, n)$ to $(M\times\mathbb{R}^n, M, \mathrm{pr}_M, n)$ (where $\mathrm{pr}_M : M\times\mathbb{R}^n \to M$). Let $\zeta : \mathrm{T}M \to M\times\mathbb{R}^n$ be such an isomorphism. Observe that the map

$$\begin{cases}\Psi_\zeta: & C^0(X\times M,\mathbb{R}^n) \to \mathrm{Dyn}(X,M)\\ & f \mapsto \Psi_\zeta(f)\end{cases},$$

with

$$\begin{cases}\Psi_\zeta(f): & X\times M \to \mathrm{T}M\\ & (x,a) \mapsto \left(\zeta^{-1}\circ(\mathrm{pr}_M^X\times f)\circ\Delta_{X\times M}\right)(x,a)\end{cases}$$

(where $\mathrm{pr}_M^X : X\times M \to M$ and $\Delta_{X\times M} : (x,a)\in X\times M \mapsto ((x,a),(x,a)) \in (X\times M)\times(X\times M)$ is the diagonal function on $X\times M$) is a bijection whose inverse is the map

$$\begin{cases}\Psi_\zeta^{-1}: & \mathrm{Dyn}(X,M) \to C^0(X\times M,\mathbb{R}^n)\\ & D \mapsto \Psi_\zeta^{-1}(D)\end{cases},$$

with

$$\begin{cases}\Psi_\zeta^{-1}(D): & X\times M \to \mathbb{R}^n\\ & (x,a) \mapsto \left(\mathrm{pr}_{\mathbb{R}^n}\circ\zeta\circ D\right)(x,a)\end{cases}$$

(where $\mathrm{pr}_{\mathbb{R}^n} : M\times\mathbb{R}^n \to \mathbb{R}^n$).

Now, we prove the first property. Let $f, f' \in C^0(X\times M,\mathbb{R}^n)$. First, let H be a homotopy between f and f'. Observe that the map

$$\begin{cases}H': & [0,1]\times X\times M \to \mathrm{T}M\\ & (t,x,a) \mapsto \Psi_\zeta(H(t,\cdot,\cdot))(x,a)\end{cases}$$

is continuous (by composition), and that $H'(0,\cdot,\cdot) = \Psi_\zeta(f)$ and $H'(1,\cdot,\cdot) = \Psi_\zeta(f')$. Furthermore, for every $t\in[0,1]$ and every $(x,a)\in X\times M$, observe that

$$\begin{aligned}(p\circ H'(t,x,\cdot))(a) &= (p\circ\zeta^{-1})(a, H(t,x,a))\\ &= \mathrm{pr}_M(a, H(t,x,a)) \text{ (by definition of } \zeta)\\ &= a,\end{aligned}$$

i.e., for every $t\in[0,1]$ and every $x\in X$, $H'(t,x,\cdot)\in\mathfrak{X}(M)$, hence that $H'(t,\cdot,\cdot)\in\mathrm{Dyn}(X,M)$. Conversely, let H be a homotopy between $\Psi_\zeta(f)$ and $\Psi_\zeta(f')$ such that for every $t\in[0,1]$, $H(t,\cdot,\cdot)\in\mathrm{Dyn}(X,M)$. Then, the map

$$\begin{cases}H': & [0,1]\times X\times M \to \mathbb{R}^n\\ & (t,x,a) \mapsto \left(\mathrm{pr}_{\mathbb{R}^n}\circ\zeta\circ H\right)(t,x,a)\end{cases}$$

is the required homotopy between f and f', since it is continuous (by composition), and since $H'(0,\cdot,\cdot) = f$ and $H'(1,\cdot,\cdot) = f'$.

Finally, we prove the second property. Let $f\in C^0(X\times M,\mathbb{R}^n)$. By definition,

$$\begin{aligned}\mathfrak{Z}(\Psi_\zeta(f)) &= \{(x,a)\in X\times M : a\in\mathfrak{Z}(\Psi_\zeta(f)(x,\cdot))\}\\ &= \{(x,a)\in X\times M : \Psi_\zeta(f)(x,a) = (a,\overrightarrow{0_a})\}\\ &= \{(x,a)\in X\times M : \zeta^{-1}(a,f(x,a)) = (a,\overrightarrow{0_a})\}\end{aligned}$$

(where $\overrightarrow{0_a}$ is the derivation at a constantly equal to 0, for every $a\in M$). First, let $(x,a)\in f^{-1}(0_{\mathbb{R}^n})$. Since ζ^{-1} is a vector bundle homomorphism, the restricted map $(a',y)\in\{a\}\times\mathbb{R}^n \mapsto \zeta^{-1}(a',y)\in\{a\}\times\mathrm{T}_aM$ is a linear map, which implies that $\zeta^{-1}(a,f(x,a)) = \zeta^{-1}(a,0) = (a,\overrightarrow{0_a})$, thus $(x,a)\in\mathfrak{Z}(\Psi_\zeta(f))$. Finally, let $(x,a)\in X\times M$ such that $\zeta^{-1}(a,f(x,a)) = (a,\overrightarrow{0_a})$. Observe that $\zeta(\zeta^{-1}(a,f(x,a))) = (a,f(x,a)) = \zeta(a,\overrightarrow{0_a})$. Since ζ is a vector bundle homomorphism, the restricted map $(a',v)\in\{a\}\times\mathrm{T}_aM \to \zeta(a',v)\in\{a\}\times\mathbb{R}^n$ is a linear map, which implies that $(a,f(x,a)) = \zeta(a,\overrightarrow{0_a}) = (a,0)$, thus $(x,a)\in f^{-1}(0_{\mathbb{R}^n})$. □

Table D.3
Table of notations for the dynamics equivalence theorem.

$\mathbf{P}_{\mathcal{V}}$	*Graph of pairwise stable networks associated with a set $\mathcal{V}$ of societies*
$\mathbf{E}_{\mathcal{V}}$	*Ambient space $\mathcal{V}\times\mathbf{G}$ associated with a set $\mathcal{V}$ of societies*
$\mathcal{V}^0$	*Trivial copy $\mathcal{V}\times\{g^0\}$ of a set $\mathcal{V}$ of societies*
$\widetilde{\mathbf{E}}_{\mathcal{V}}$	*ε-ambient space $\mathcal{V}\times\widetilde{\mathbf{G}}$ associated with a set $\mathcal{V}$ of societies*
$\kappa^0_{\mathcal{V}} : \mathcal{V}\to\widetilde{\mathbf{E}}_{\mathcal{V}}$	*Unknot associated with $\mathcal{V}$ (and g^0), where $\mathcal{V}$ is a set of societies*
$\kappa_{\mathcal{V}} : \mathcal{V}\to\widetilde{\mathbf{E}}_{\mathcal{V}}$	*Knot of pairwise stable networks associated with $\mathcal{V}$ (and $\rho_{\mathcal{V}}$), where $\mathcal{V}$ is a regular set of societies*
$\theta_{\mathcal{V}} : [0,1]\times\widetilde{\mathbf{E}}_{\mathcal{V}}\to\widetilde{\mathbf{E}}_{\mathcal{V}}$	*Ambient isotopy between $\kappa_{\mathcal{V}}$ and $\kappa^0_{\mathcal{V}}$ of the unknotteddness theorem, where $\mathcal{V}$ is a regular set of societies*

C.2. Proof of Corollary 6.1

For every vector bundle isomorphism $\zeta : \mathrm{T}M \to M\times\mathbb{R}^n$, following the constructions of the proof of Proposition 6.1, the map

$$\begin{cases} \Psi_\zeta : \mathcal{C}^0(M,\mathbb{R}^n) \to \mathfrak{X}(M) \\ \quad f \mapsto \begin{cases} \Psi_\zeta(f) : M \to \mathrm{T}M \\ \quad a \mapsto \left(\zeta^{-1}\circ(\mathrm{id}_M\times f)\circ\Delta_M\right)(a) \end{cases} \end{cases}$$

is a bijection whose inverse is the map

$$\begin{cases} \Psi_\zeta^{-1} : \mathfrak{X}(M) \to \mathcal{C}^0(M,\mathbb{R}^n) \\ \quad X \mapsto \begin{cases} \Psi_\zeta^{-1}(X) : M \to \mathbb{R}^n \\ \quad a \mapsto \left(\mathrm{pr}_{\mathbb{R}^n}\circ\zeta\circ X\right)(a) \end{cases} \end{cases}$$

and satisfies the two required properties. □

Appendix D. Proof of the dynamics equivalence theorem

We provide in Table D.3 a list of the notations used throughout the proof.

Consider a fixed regular set $\mathcal{V}$ of societies (recall that $\varepsilon > 0$ and $g^0 \in \mathbf{G}$ are fixed). Also, consider two strongly inward-pointing extended network dynamics $\widetilde{D}, \widetilde{D}' : \widetilde{\mathbf{E}}_{\mathcal{V}} \to \mathbb{R}^\ell$ on $\mathcal{V}$ (as explained at the beginning of Section 6, any extended network dynamics $\widetilde{D}$ on $\mathcal{V}$ can be treated as a continuous map from $\widetilde{\mathbf{E}}_{\mathcal{V}}$ to $\mathbb{R}^\ell$ such that $\widetilde{D}^{-1}(0_{\mathbb{R}^\ell}) = \mathbf{P}_{\mathcal{V}}$; this identification is used throughout the proof for simplicity). Recall that $\theta_{\mathcal{V}} : \widetilde{\mathbf{E}}_{\mathcal{V}} : [0,1]\times\widetilde{\mathbf{E}}_{\mathcal{V}} \to \widetilde{\mathbf{E}}_{\mathcal{V}}$ denotes the ambient isotopy between the knot $\kappa_{\mathcal{V}} : \mathcal{V} \to \widetilde{\mathbf{E}}_{\mathcal{V}}$ of pairwise stable networks associated with $\mathcal{V}$ and the unknot $\kappa^0_{\mathcal{V}} : \mathcal{V} \to \widetilde{\mathbf{E}}_{\mathcal{V}}$ associated with $\mathcal{V}$ in the unknotteddness theorem (see Section 5). Finally, the following lemma will be used multiple times in this proof.

Lemma D.1. *Let X, Y be two metrizable topological spaces, with A being a subspace of X, and let $f_1 : A \to Y$ and $f_2 : X\backslash A \to Y$ be continuous functions. Suppose that the following conditions hold:*

1. *For every $x^* \in \partial A$ and every sequence $(x^n)_{n\in\mathbb{N}}$ in A such that $x^n \xrightarrow{n\to+\infty} x^*$, we have $f_1(x^n) \xrightarrow{n\to+\infty} f(x^*)$*
2. *For every $x^* \in \partial(X\backslash A)$ and every sequence $(x^n)_{n\in\mathbb{N}}$ in $X\backslash A$ such that $x^n \xrightarrow{n\to+\infty} x^*$, we have $f_2(x^n) \xrightarrow{n\to+\infty} f(x^*)$.*

Then, the map

$$\begin{cases} f : X \to Y \\ \quad x \mapsto \begin{cases} f_1(x) & \text{if} \quad x\in A \\ f_2(x) & \text{if} \quad x\in X\backslash A \end{cases} \end{cases}$$

is continuous.

Proof. Let $x^* \in X$ and $(x^n)_{n\in\mathbb{N}}$ be a sequence in X such that $x^n \xrightarrow{n\to+\infty} x^*$. First, suppose that $x^* \in \mathrm{int}(A)$ (in that case, $f(x^*) = f_1(x^*)$). Since $x^n \xrightarrow{n\to+\infty} x^*$, there exists $m \in \mathbb{N}$ such that for every $n \geq m$, $x^n \in \mathrm{int}(A)$ (which implies that $f(x^n) = f_1(x^n)$ for every $n \geq m$). Hence, by continuity of f_1, $f(x^n) \xrightarrow{n\to+\infty} f(x^*)$, which means that f is continuous at x^*. Similarly, if $x^* \in \mathrm{int}(X\backslash A)$ (in that case, $f(x^*) = f_2(x^*)$), then by continuity of f_2, f is continuous at x^*. Lastly, suppose that $x^* \in \partial A = \partial(X\backslash A)$. Consider the sets of indices $I_A = \{n\in\mathbb{N} : x^n \in A\}$ and $I_{X\backslash A} = \{n\in\mathbb{N} : x^n \in X\backslash A\}$, and let $(x^n_A)_{n\in\mathbb{N}} = (x^n)_{n\in I_A}$ be the subsequence in A, and $(x^n_{X\backslash A})_{n\in\mathbb{N}} = (x^n)_{n\in I_{X\backslash A}}$ be the subsequence in $X\backslash A$. Since $x^n \xrightarrow{n\to+\infty} x^*$, it follows that $x^n_A \xrightarrow{n\to+\infty} x^*$ and $x^n_{X\backslash A} \xrightarrow{n\to+\infty} x^*$. Thus by the two assumptions of the lemma, we have $f_1(x^n_A) \xrightarrow{n\to+\infty} f(x^*)$ and $f_2(x^n_{X\backslash A}) \xrightarrow{n\to+\infty} f(x^*)$. In other words, for any open subset $V \subset Y$ containing $f(x^*)$, there exist $m_A \in \mathbb{N}$ and $m_{X\backslash A} \in \mathbb{N}$ such that for every $n \geq m_A$, $f_1(x^n_A) \in V$, and for every $n \geq m_{X\backslash A}$, $f_2(x^n_{X\backslash A}) \in V$. Finally, by the definition of f and since $\mathbb{N} = I_A \cup I_{X\backslash A}$ with $I_A \cap I_{X\backslash A} = \emptyset$, we observe that for every $n \in \mathbb{N}$, $f(x^n)$ is either equal to $f_1(x^{m(n)}_A)$ or to $f_2(x^{m(n)}_{X\backslash A})$, for some $m(n) \in \mathbb{N}$. Therefore, setting $m = m_A + m_{X\backslash A}$, we obtain that for every $n \geq m$, $f(x^n) \in V$, meaning that f is continuous at x^*. □

Step I. Construction of a homotopy $\widetilde{H}_{\mathcal{V}}$ between $\widetilde{D}$ and $\widetilde{D}'$ (1). First, define

$$\begin{cases} \widetilde{D}_\partial : \widetilde{\mathbf{E}}_{\mathcal{V}} \to \mathbb{R}^\ell \\ \quad (v,g) \mapsto \begin{cases} \tau^g\widetilde{D}(v,g) + (1-\tau^g)\widetilde{D}'(v,g) & \text{if} \quad g\in\widetilde{\mathbf{G}}\backslash\mathbf{G} \\ \widetilde{D}(v,g) & \text{otherwise} \end{cases} \end{cases},$$

where for every $g \in \widetilde{\mathbf{G}}$,

$$\tau^g = \frac{\|g - g^{1/2}\|_\infty - \frac{1}{2} - \varepsilon}{-\varepsilon},$$

with $g^{1/2} \in \mathbf{G}$ being the network such that for every $ij \in \mathbf{L}$, $g^{1/2}_{ij} = \frac{1}{2}$. The map $\widetilde{D}_\partial$ is continuous because: (i) the map $(v,g) \in \mathcal{V}\times\mathbf{G}\backslash\mathbf{G} \mapsto \tau^g\widetilde{D}(v,g) + (1-\tau^g)\widetilde{D}'(v,g) \in \mathbb{R}^\ell$ is continuous (in particular, the map $g \in \widetilde{\mathbf{G}}\backslash\mathbf{G} \mapsto \tau^g \in \mathbb{R}$ is continuous by itself); (ii) the map $(v,g) \in \mathcal{V}\times\mathrm{int}(\mathbf{G}) \mapsto \widetilde{D}(v,g) \in \mathbb{R}^\ell$ is continuous as the restriction of the continuous map $\widetilde{D}$ from $\mathcal{V}\times\mathrm{int}(\mathbf{G}) \subset \widetilde{\mathbf{E}}_{\mathcal{V}}$ to $\mathbb{R}^\ell$; (iii) for every $(v,g) \in \mathcal{V}\times\partial\mathbf{G}$, we have $\tau^g\widetilde{D}(v,g) + (1-\tau^g)\widetilde{D}'(v,g) = \widetilde{D}(v,g)$. Moreover, notice that

$$\widetilde{D}_\partial|_{\mathbf{E}_{\mathcal{V}}} = \widetilde{D}|_{\mathbf{E}_{\mathcal{V}}} \text{ and } \widetilde{D}_\partial|_{\mathcal{V}\times\partial\widetilde{\mathbf{G}}} = \widetilde{D}'|_{\mathcal{V}\times\partial\widetilde{\mathbf{G}}}. \tag{D.1}$$

Next, define

$$\begin{cases} \widetilde{H}^1_{\mathcal{V}} : [0,1]\times\widetilde{\mathbf{E}}_{\mathcal{V}} \to \mathbb{R}^\ell \\ \quad (t,(v,g)) \mapsto (1-t)\widetilde{D}(v,g) + t\widetilde{D}_\partial(v,g) \end{cases}$$

The map $\widetilde{H}^1_{\mathcal{V}}$ is a homotopy between $\widetilde{D}$ and $\widetilde{D}_\partial$; more precisely, it corresponds to the straight-line homotopy between $\widetilde{D}$ and $\widetilde{D}_\partial$. Furthermore, for every $t \in [0,1]$, the map $\widetilde{H}^1_{\mathcal{V}}(t,\cdot,\cdot)$ defines an extended network dynamics on $\mathcal{V}$. Indeed, for every $t \in [0,1]$: (i) $\widetilde{H}^1_{\mathcal{V}}(t,\cdot,\cdot) : \widetilde{\mathbf{E}}_{\mathcal{V}} \to \mathbb{R}^\ell$ is continuous because $\widetilde{H}^1_{\mathcal{V}}$ is itself a continuous map; (ii) $(\widetilde{H}^1_{\mathcal{V}}(t,\cdot,\cdot))^{-1}(0_{\mathbb{R}^\ell}) = \mathbf{P}_{\mathcal{V}}$. To understand the last point, observe that for every $v \in \mathcal{V}$, every $\varepsilon' \in (0,\varepsilon]$, every $g \in \partial\mathbf{G}^{\varepsilon'}$, and every $x \in \mathrm{N}_{\mathbf{G}^{\varepsilon'}}(g)\backslash\{0_{\mathbb{R}^\ell}\}$, we have:

$$\begin{aligned} \langle\widetilde{D}_\partial(v,g),x\rangle &= \langle\tau^g\widetilde{D}(v,g) + (1-\tau^g)\widetilde{D}'(v,g),x\rangle \\ &\quad (\text{since } \partial\mathbf{G}^{\varepsilon'} \subset \widetilde{\mathbf{G}}\backslash\mathbf{G}, \text{ for every } \varepsilon' \in (0,\varepsilon)) \\ &= \tau^g\langle\widetilde{D}(v,g),x\rangle + (1-\tau^g)\langle\widetilde{D}'(v,g),x\rangle < 0 \end{aligned}$$

since τ^g belongs to $[0,1]$ (as g belongs to $\widetilde{\mathbf{G}}\backslash\mathbf{G}$), and since both $\widetilde{D}$ and $\widetilde{D}'$ are strongly inward-pointing (by assumption). Hence, for every $v \in \mathcal{V}$, $(\widetilde{D}_\partial)_v$ has no zeros on $\widetilde{\mathbf{G}}\backslash\mathbf{G}$. Moreover, knowing from Eq. (D.1) that $\widetilde{D}_\partial|_{\mathbf{E}_{\mathcal{V}}} = \widetilde{D}|_{\mathbf{E}_{\mathcal{V}}}$, one immediately obtains that $(\widetilde{D}_\partial)^{-1}(0_{\mathbb{R}^\ell}) = \mathbf{P}_{\mathcal{V}}$. Finally, observe that for every $v \in \mathcal{V}$, every $\varepsilon' \in (0,\varepsilon]$, every $g \in \partial\mathbf{G}^{\varepsilon'}$, every $x \in \mathrm{N}_{\mathbf{G}^{\varepsilon'}}(g)\backslash\{0_{\mathbb{R}^L}\}$, and every $t \in [0,1]$, we have:

$$\begin{aligned} \langle\widetilde{H}^1_{\mathcal{V}}(t,v,g),x\rangle &= \langle(1-t)\widetilde{D}(v,g) + t\widetilde{D}_\partial(v,g),x\rangle \\ &= (1-t)\langle\widetilde{D}(v,g),x\rangle + t\langle\widetilde{D}_\partial(v,g),x\rangle < 0 \end{aligned}$$

since, by assumption, $\widetilde{D}$ is strongly inward-pointing, and from the previous calculation, $\widetilde{D}_\partial$ is also strongly inward-pointing. Hence, for

every $v \in \mathcal{V}$, $\widetilde{H}^1_{\mathcal{V}}(t,v,\cdot)$ has no zeros on $\widetilde{\mathbf{G}}\backslash\mathbf{G}$. Therefore, it follows that for every $t \in [0,1]$, $(\widetilde{H}^1_{\mathcal{V}}(t,\cdot,\cdot))^{-1}(0_{\mathbb{R}^\ell}) = \mathbf{P}_{\mathcal{V}}$, since for every $(v,g) \in \mathbf{E}_{\mathcal{V}}$, we have:

$$\begin{aligned}\widetilde{H}^1_{\mathcal{V}}(t,v,g) &= (1-t)\widetilde{D}(v,g) + t\widetilde{D}_\partial(v,g)\\ &= (1-t)\widetilde{D}(v,g) + t\widetilde{D}(v,g) \text{ (from Eq. (D.1))}\\ &= \widetilde{D}(v,g).\end{aligned}$$

Step II. Use of the ambient isotopy $\theta_{\mathcal{V}}$ from the unknottedness theorem. Define

$$\widetilde{\Delta}_\partial = \widetilde{D}_\partial \circ (\theta_{\mathcal{V}}(1,\cdot,\cdot))^{-1} \text{ and } \widetilde{\Delta}' = \widetilde{D}' \circ (\theta_{\mathcal{V}}(1,\cdot,\cdot))^{-1}.$$

The map $\widetilde{\Delta}_\partial$ (resp. $\widetilde{\Delta}'$) is continuous, as it is the composition of two continuous maps, and satisfies

$$\begin{aligned}\widetilde{\Delta}_\partial^{-1}(0_{\mathbb{R}^L}) &= \left(\widetilde{D}_\partial \circ (\theta_{\mathcal{V}}(1,\cdot,\cdot))^{-1}\right)^{-1}(0_{\mathbb{R}^\ell})\\ &= \theta_{\mathcal{V}}(1,\cdot,\cdot)\left(\widetilde{D}_\partial^{-1}(0_{\mathbb{R}^\ell})\right)\\ &= \theta_{\mathcal{V}}(1,\cdot,\cdot)(\mathbf{P}_{\mathcal{V}}) = \mathcal{V}^0\end{aligned}$$

(resp. $(\widetilde{\Delta}')^{-1}(0_{\mathbb{R}^\ell}) = \mathcal{V}^0$). Now, suppose that there exists a homotopy $\widetilde{\Phi}_{\mathcal{V}}$ between $\widetilde{\Delta}_\partial$ and $\widetilde{\Delta}'$ such that for every $t \in [0,1]$, $(\widetilde{\Phi}_{\mathcal{V}}(t,\cdot,\cdot))^{-1}(0_{\mathbb{R}^\ell}) = \mathcal{V}^0$. Then, defining

$$\begin{cases}\widetilde{H}^2_{\mathcal{V}}: & [0,1]\times\widetilde{\mathbf{E}}_{\mathcal{V}} \to \mathbb{R}^\ell\\ & (t,(v,g)) \mapsto (\widetilde{\Phi}_{\mathcal{V}}(t,\cdot,\cdot)\circ\theta_{\mathcal{V}}(1,\cdot,\cdot))(v,g)\end{cases},$$

one obtains a homotopy between $\widetilde{D}_\partial$ and $\widetilde{D}'$. Moreover, for every $t \in [0,1]$, the map $\widetilde{H}^2_{\mathcal{V}}(t,\cdot,\cdot)$ defines an extended network dynamics on $\mathcal{V}$.

Finally, recall that the ambient isotopy $\theta_{\mathcal{V}}$ does not deform the boundary of the ambient space $\widetilde{\mathbf{E}}_{\mathcal{V}}$, i.e., for every $t \in [0,1]$, $\theta_{\mathcal{V}}(t,\cdot,\cdot)|_{\mathcal{V}\times\partial\widetilde{\mathbf{G}}} = \mathrm{id}_{\mathcal{V}\times\partial\widetilde{\mathbf{G}}}$. Thus, from Eq. (D.1), it follows that

$$\widetilde{\Delta}_\partial|_{\mathcal{V}\times\partial\widetilde{\mathbf{G}}} = \widetilde{\Delta}'|_{\mathcal{V}\times\partial\widetilde{\mathbf{G}}}. \tag{D.2}$$

Step III. Construction of a homotopy $\widetilde{\Phi}_{\mathcal{V}}$ between $\widetilde{\Delta}_\partial$ and $\widetilde{\Delta}'$. Substep III.1

First, define the function

$$\begin{cases}\widetilde{\Delta}^1: & \widetilde{\mathbf{E}}_{\mathcal{V}} \to \mathbb{R}^\ell\\ & (v,g) \mapsto \begin{cases}\frac{\widetilde{\Delta}_\partial(v,g)}{\|\widetilde{\Delta}_\partial(v,g)\|_2}\|g-g^0\|_2 & \text{if } g \neq g^0\\ 0_{\mathbb{R}^\ell} & \text{otherwise}\end{cases}\end{cases}.$$

The continuity of $\widetilde{\Delta}^1$ follows from Lemma D.1 applied to $X = \widetilde{\mathbf{E}}_{\mathcal{V}}$, $Y = \mathbb{R}^\ell$, $A = \mathcal{V}\times(\widetilde{\mathbf{G}}\backslash\{g^0\})$, $f_1 : (v,g) \in A \mapsto \frac{\widetilde{\Delta}_\partial(v,g)}{\|\widetilde{\Delta}_\partial(v,g)\|_2}\|g-g^0\|_2 \in Y$ and $f_2 : (v,g) \in X\backslash A \mapsto 0_{\mathbb{R}^\ell} \in Y$. The second assumption of Lemma D.1 is clearly satisfied. To verify the first one, consider a point $(v^*,g^*) \in \partial A = \mathcal{V}^0$ and a sequence $(v^n,g^n)_{\ell\in\mathbb{N}}$ in A such that $(v^n,g^n) \xrightarrow{n\to+\infty} (v^*,g^*)$. For every $n \in \mathbb{N}$, we have:

$$\begin{aligned}\|f_1(v^n,g^n) - \widetilde{\Delta}^1(v^*,g^*)\|_2 &= \|\left(\frac{\widetilde{\Delta}_\partial(v^n,g^n)}{\|\widetilde{\Delta}_\partial(v^n,g^n)\|_2}\|g^n-g^0\|_2\right) - 0_{\mathbb{R}^\ell}\|_2\\ &\quad \text{(since } g^* = g^0)\\ &= \|\frac{\widetilde{\Delta}_\partial(v^n,g^n)}{\|\widetilde{\Delta}_\partial(v^n,g^n)\|_2}\|g^n-g^0\|_2\|_2\\ &= \frac{\|\widetilde{\Delta}_\partial(v^n,g^n)\|_2}{\|\widetilde{\Delta}_\partial(v^n,g^n)\|_2}\|g^n-g^0\|_2\\ &= \|g^n-g^0\|_2.\end{aligned}$$

Since $g^n \xrightarrow{n\to+\infty} g^*$ and $g^* = g^0$, it follows that $\|f_1(v^n,g^n) - \widetilde{\Delta}^1(v^*,g^*)\|_2 \xrightarrow{n\to+\infty} 0$. Thus that $f_1(v^n,g^n) \xrightarrow{n\to+\infty} \widetilde{\Delta}^1(v^*,g^*)$, proving the continuity of $\widetilde{\Delta}^1$.

Next, define

$$\begin{cases}\widetilde{\Phi}^1_{\mathcal{V}}: & [0,1]\times\widetilde{\mathbf{E}}_{\mathcal{V}} \to \mathbb{R}^\ell\\ & (t,(v,g)) \mapsto (1-t)\widetilde{\Delta}_\partial(v,g) + t\widetilde{\Delta}^1(v,g)\end{cases}.$$

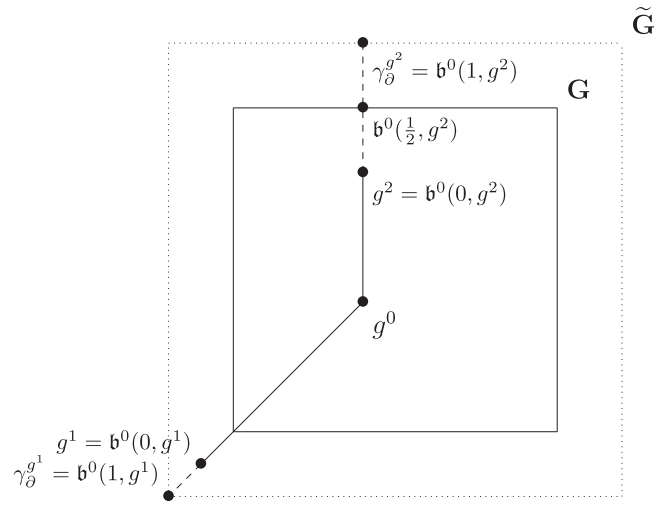

Fig. D.6. $\mathfrak{b}^0$ *considering two networks* g^1 *and* g^2.

This function provides a straight-line homotopy between $\widetilde{\Delta}_\partial$ and $\widetilde{\Delta}^1$. For every $t \in [0,1]$, we verify that $(\widetilde{\Phi}^1_{\mathcal{V}}(t,\cdot,\cdot))^{-1}(0_{\mathbb{R}^\ell}) = \mathcal{V}^0$. Indeed, a point $(v,g) \in \widetilde{\mathbf{E}}_{\mathcal{V}}$ belongs to $(\widetilde{\Phi}^1_{\mathcal{V}}(t,\cdot,\cdot))^{-1}(0_{\mathbb{R}^\ell})$ if and only if: (i) either $g = g^0$; (ii) or $g \neq g^0$ and

$$(1-t)\widetilde{\Delta}_\partial(v,g) + t\left(\frac{\widetilde{\Delta}_\partial(v,g)}{\|\widetilde{\Delta}_\partial(v,g)\|_2}\|g-g^0\|_2\right) = 0_{\mathbb{R}^\ell}. \tag{D.3}$$

However, Eq. (D.3) has no solution if $g \neq g^0$ because $\widetilde{\Delta}_\partial^{-1}(0_{\mathbb{R}^\ell}) = \mathcal{V}^0$. Thus, for every $t \in [0,1]$, $(\widetilde{\Phi}^1_{\mathcal{V}}(t,\cdot,\cdot))^{-1}(0_{\mathbb{R}^\ell}) = \mathcal{V}^0$.

Substep III.2. For every $g \in \widetilde{\mathbf{G}}\backslash\{g^0\}$, consider the unique element $\gamma_\partial^g \in \partial\widetilde{\mathbf{G}}$ such that $\gamma_\partial^g \in \{\gamma \in \widetilde{\mathbf{G}} : \exists t \in [0,+\infty), \gamma = (1-t)g^0 + tg\}$, i.e., such that γ_∂^g belongs to the half-line starting at g^0 and passing through g. Moreover, consider the map

$$\begin{cases}\mathfrak{b}^0: & [0,1]\times(\widetilde{\mathbf{G}}\backslash\{g^0\}) \to \widetilde{\mathbf{G}}\\ & (t,g) \mapsto (1-t)g + t\gamma_\partial^g\end{cases}.$$

This map is continuous, as the function $g \in \widetilde{\mathbf{G}}\backslash\{g^0\} \mapsto \gamma_\partial^g \in \partial\widetilde{\mathbf{G}}$ is itself continuous. Fig. D.6 illustrates the behavior $\mathfrak{b}^0$.

Now, define

$$\begin{cases}\widetilde{\Delta}^2: & \widetilde{\mathbf{E}}_{\mathcal{V}} \to \mathbb{R}^\ell\\ & (v,g) \mapsto \begin{cases}\frac{\widetilde{\Delta}_\partial(v,\gamma_\partial^g)}{\|\widetilde{\Delta}_\partial(v,\gamma_\partial^g)\|_2}\|g-g^0\|_2 & \text{if } g \neq g^0\\ 0_{\mathbb{R}^\ell} & \text{otherwise}\end{cases}\end{cases}.$$

The continuity of this map is a consequence of Lemma D.1 applied to $X = \widetilde{\mathbf{E}}_{\mathcal{V}}$, $Y = \mathbb{R}^\ell$, $A = \mathcal{V}\times(\widetilde{\mathbf{G}}\backslash\{g^0\})$, $f_1 : (v,g) \in A \mapsto \frac{\widetilde{\Delta}_\partial(v,\gamma_\partial^g)}{\|\widetilde{\Delta}_\partial(v,\gamma_\partial^g)\|_2}\|g-g^0\|_2 \in Y$ and $f_2 : (v,g) \in X\backslash A \mapsto 0_{\mathbb{R}^\ell} \in Y$. The second assumption of Lemma D.1 is clearly verified. To verify the first one, consider a point $(v^*,g^*) \in \partial A = \mathcal{V}^0$ and a sequence $(v^n,g^n)_{n\in\mathbb{N}}$ in A such that $(v^n,g^n) \xrightarrow{n\to+\infty} (v^*,g^*)$. For every $n \in \mathbb{N}$,

$$\begin{aligned}\|f_1(v^n,g^n) - \widetilde{\Delta}^2(v^*,g^*)\|_2 &= \|\left(\frac{\widetilde{\Delta}_\partial(v^n,\gamma_\partial^{g^n})}{\|\widetilde{\Delta}_\partial(v,\gamma_\partial^{g^n})\|_2}\|g^n-g^0\|_2\right) - 0_{\mathbb{R}^\ell}\|_2\\ &\quad \text{(since } g^* = g^0)\\ &= \|\frac{\widetilde{\Delta}_\partial(v^n,\gamma_\partial^{g^n})}{\|\widetilde{\Delta}_\partial(v^n,\gamma_\partial^{g^n})\|_2}\|g^n-g^0\|_2\|_2\\ &= \frac{\|\widetilde{\Delta}_\partial(v^n,\gamma_\partial^{g^n})\|_2}{\|\widetilde{\Delta}_\partial(v^n,\gamma_\partial^{g^n})\|_2}\|g^n-g^0\|_2\\ &= \|g^n-g^0\|_2.\end{aligned}$$

Since $g^n \xrightarrow{n\to+\infty} g^*$ and $g^* = g^0$, it follows that $\|f_1(v^n, g^n) - \widetilde{\Delta}^2(v^*, g^*)\|_2 \xrightarrow{n\to+\infty} 0$. Thus that $f_1(v^n, g^n) \xrightarrow{n\to+\infty} \widetilde{\Delta}^2(v^*, g^*)$, proving the continuity of $\widetilde{\Delta}^2$.

Next, define

$$\begin{cases} \widetilde{\Phi}^2_{\mathcal{V}} : [0,1]\times\widetilde{\mathbf{E}}_{\mathcal{V}} \to \mathbb{R}^\ell \\ (t,(v,g)) \mapsto \begin{cases} \frac{\widetilde{\Delta}_\partial(v,\mathfrak{b}^0(t,g))}{\|\widetilde{\Delta}_\partial(v,\mathfrak{b}^0(t,g))\|_2}\|g-g^0\|_2 & \text{if} \quad g\neq g^0 \\ 0_{\mathbb{R}^\ell} & \text{otherwise} \end{cases} \end{cases}.$$

This map is a homotopy between $\widetilde{\Delta}^1$ and $\widetilde{\Delta}^2$:

1. Again, the continuity of this map follows from Lemma D.1 applied to $X = [0,1]\times\widetilde{\mathbf{E}}_{\mathcal{V}}$, $Y = \mathbb{R}^\ell$, $A = [0,1]\times\left(\mathcal{V}\times(\widetilde{\mathbf{G}}\backslash\{g^0\})\right)$, $f_1 : (t,(v,g)) \in A \mapsto \frac{\widetilde{\Delta}_\partial(v,\mathfrak{b}^0(t,g))}{\|\widetilde{\Delta}_\partial(v,\mathfrak{b}^0(t,g))\|_2}\|g-g^0\|_2 \in Y$ and $f_2 : (t,(v,g)) \in X\backslash A \mapsto 0_{\mathbb{R}^\ell} \in Y$.
2. For every $(v,g)\in\widetilde{\mathbf{E}}_{\mathcal{V}}$,
$$\begin{aligned}\widetilde{\Phi}^2_{\mathcal{V}}(0,v,g) &= \begin{cases} \frac{\widetilde{\Delta}_\partial(v,\mathfrak{b}^0(0,g))}{\|\widetilde{\Delta}_\partial(v,\mathfrak{b}^0(0,g))\|_2}\|g-g^0\|_2 & \text{if} \quad g\neq g^0 \\ 0_{\mathbb{R}^\ell} & \text{otherwise}\end{cases} \\ &= \begin{cases} \frac{\widetilde{\Delta}_\partial(v,g)}{\|\widetilde{\Delta}_\partial(v,g)\|_2}\|g-g^0\|_2 & \text{if} \quad g\neq g^0 \\ 0_{\mathbb{R}^\ell} & \text{otherwise}\end{cases} \\ &= \widetilde{\Delta}^1(v,g).\end{aligned}$$
3. For every $(v,g)\in\widetilde{\mathbf{E}}_{\mathcal{V}}$,
$$\begin{aligned}\widetilde{\Phi}^2_{\mathcal{V}}(1,v,g) &= \begin{cases} \frac{\widetilde{\Delta}_\partial(v,\mathfrak{b}^0(1,g))}{\|\widetilde{\Delta}_\partial(v,\mathfrak{b}^0(1,g))\|_2}\|g-g^0\|_2 & \text{if} \quad g\neq g^0 \\ 0_{\mathbb{R}^\ell} & \text{otherwise}\end{cases} \\ &= \begin{cases} \frac{\widetilde{\Delta}_\partial(v,\gamma^g_\partial)}{\|\widetilde{\Delta}_\partial(v,\gamma^g_\partial)\|_2}\|g-g^0\|_2 & \text{if} \quad g\neq g^0 \\ 0_{\mathbb{R}^\ell} & \text{otherwise}\end{cases} \\ &= \widetilde{\Delta}^2(v,g).\end{aligned}$$

For every $t\in[0,1]$, we verify that $(\widetilde{\Phi}^2_{\mathcal{V}}(t,\cdot,\cdot))^{-1}(0_{\mathbb{R}^\ell}) = \mathcal{V}^0$. Indeed, a point $(v,g)\in\widetilde{\mathbf{E}}_{\mathcal{V}}$ belongs to $(\widetilde{\Phi}^2_{\mathcal{V}}(t,\cdot,\cdot))^{-1}(0_{\mathbb{R}^\ell})$ if and only if: (i) either $g = g^0$; (ii) or $g\neq g^0$ and

$$\frac{\widetilde{\Delta}_\partial(v,\mathfrak{b}^0(t,g))}{\|\widetilde{\Delta}_\partial(v,\mathfrak{b}^0(t,g))\|_2}\|g-g^0\|_2 = 0_{\mathbb{R}^\ell} \tag{D.4}$$

However, Eq. (D.4) has no solution if $g\neq g^0$ because $(\widetilde{\Delta}_\partial)^{-1}(0_{\mathbb{R}^\ell}) = \mathcal{V}^0$. Thus, for every $t\in[0,1]$, $(\widetilde{\Phi}^2_{\mathcal{V}}(t,\cdot,\cdot))^{-1}(0_{\mathbb{R}^\ell}) = \mathcal{V}^0$.

Substep III.3. First, observe that the maps $\widehat{\Delta}^2$ (defined in Substep III.2) and

$$\begin{cases} \widehat{\Delta}^2 : \widetilde{\mathbf{E}}_{\mathcal{V}} \to \mathbb{R}^\ell \\ (v,g) \mapsto \begin{cases} \frac{\widetilde{\Delta}'(v,\gamma^g_\partial)}{\|\widetilde{\Delta}'(v,\gamma^g_\partial)\|_2}\|g-g^0\|_2 & \text{if} \quad g\neq g^0 \\ 0_{\mathbb{R}^\ell} & \text{otherwise}\end{cases}\end{cases}$$

are equal. Indeed, recall from Eq. (D.2) of Step II that $\widetilde{\Delta}_\partial|_{\mathcal{V}\times\partial\widetilde{\mathbf{G}}} = \widetilde{\Delta}'|_{\mathcal{V}\times\partial\widetilde{\mathbf{G}}}$ and that for every $g\in\mathbf{G}\backslash\{g^0\}$, γ^g_∂ belongs to $\partial\widetilde{\mathbf{G}}$.

Next, define the map

$$\begin{cases} \widehat{\Delta}^1 : \widetilde{\mathbf{E}}_{\mathcal{V}} \to \mathbb{R}^\ell \\ (v,g) \mapsto \begin{cases} \frac{\widetilde{\Delta}'(v,g)}{\|\widetilde{\Delta}'(v,g)\|_2}\|g-g^0\|_2 & \text{if} \quad g\neq g^0 \\ 0_{\mathbb{R}^\ell} & \text{otherwise}\end{cases}\end{cases}.$$

This map is continuous, and its proof follows similarly to the continuity proof of $\widetilde{\Delta}^1$ in Substep III.1.

Last, define the map

$$\begin{cases} \widetilde{\Phi}^3_{\mathcal{V}} : [0,1]\times\widetilde{\mathbf{E}}_{\mathcal{V}} \to \mathbb{R}^\ell \\ (t,(v,g)) \mapsto \begin{cases} \frac{\widetilde{\Delta}'(v,\mathfrak{b}^0(1-t,g))}{\|\widetilde{\Delta}'(v,\mathfrak{b}^0(1-t,g))\|_2}\|g-g^0\|_2 & \text{if} \quad g\neq g^0 \\ 0_{\mathbb{R}^\ell} & \text{otherwise}\end{cases}\end{cases}$$

(recall that the function $\mathfrak{b}^0$ is defined at the beginning of Substep III.2). This map is a homotopy between $\widehat{\Delta}^2$ and $\widehat{\Delta}^1$:

1. The proof of the continuity of $\widetilde{\Phi}^3_{\mathcal{V}}$ follows similarly to that of $\widetilde{\Phi}^2_{\mathcal{V}}$ in Substep III.2.
2. For every $(v,g)\in\widetilde{\mathbf{E}}_{\mathcal{V}}$,
$$\begin{aligned}\widetilde{\Phi}^3_{\mathcal{V}}(0,v,g) &= \begin{cases} \frac{\widetilde{\Delta}'(v,\mathfrak{b}^0(1,g))}{\|\widetilde{\Delta}'(v,\mathfrak{b}^0(1,g))\|_2}\|g-g^0\|_2 & \text{if} \quad g\neq g^0 \\ 0_{\mathbb{R}^\ell} & \text{otherwise}\end{cases} \\ &= \begin{cases} \frac{\widetilde{\Delta}'(v,\gamma^g_\partial)}{\|\widetilde{\Delta}'(v,\gamma^g_\partial)\|_2}\|g-g^0\|_2 & \text{if} \quad g\neq g^0 \\ 0_{\mathbb{R}^\ell} & \text{otherwise}\end{cases} \\ &= \widehat{\Delta}^2(v,g).\end{aligned}$$
3. For every $(v,g)\in\widetilde{\mathbf{E}}_{\mathcal{V}}$,
$$\begin{aligned}\widetilde{\Phi}^3_{\mathcal{V}}(1,v,g) &= \begin{cases} \frac{\widetilde{\Delta}'(v,\mathfrak{b}^0(0,g))}{\|\widetilde{\Delta}'(v,\mathfrak{b}^0(0,g))\|_2}\|g-g^0\|_2 & \text{if} \quad g\neq g^0 \\ 0_{\mathbb{R}^\ell} & \text{otherwise}\end{cases} \\ &= \begin{cases} \frac{\widetilde{\Delta}'(v,g)}{\|\widetilde{\Delta}'(v,g)\|_2}\|g-g^0\|_2 & \text{if} \quad g\neq g^0 \\ 0_{\mathbb{R}^\ell} & \text{otherwise}\end{cases} \\ &= \widehat{\Delta}^1(v,g).\end{aligned}$$

Furthermore, for every $t\in[0,1]$, observe that $(\widetilde{\Phi}^3_{\mathcal{V}}(t,\cdot,\cdot))^{-1}(0_{\mathbb{R}^\ell}) = \mathcal{V}^0$. The proof is analogous to that of $(\widetilde{\Phi}^2_{\mathcal{V}}(t,\cdot,\cdot))^{-1}(0_{\mathbb{R}^\ell}) = \mathcal{V}^0$, as shown in Substep III.2.

Substep III.4. Define the map

$$\begin{cases} \widetilde{\Phi}^4_{\mathcal{V}} : [0,1]\times\widetilde{\mathbf{E}}_{\mathcal{V}} \to \mathbb{R}^\ell \\ (t,(v,g)) \mapsto (1-t)\widehat{\Delta}^1(v,g) + t\widetilde{\Delta}'(v,g)\end{cases}.$$

This map defines a straight-line homotopy between $\widehat{\Delta}^1$ and $\widetilde{\Delta}'$. Furthermore, for every $t\in[0,1]$, observe that $(\widetilde{\Phi}^4_{\mathcal{V}}(t,\cdot,\cdot))^{-1}(0_{\mathbb{R}^\ell}) = \mathcal{V}^0$. The proof is analogous to that of $(\widetilde{\Phi}^1_{\mathcal{V}}(t,\cdot,\cdot))^{-1}(0_{\mathbb{R}^\ell}) = \mathcal{V}^0$, as shown in Substep III.1.

Substep III.5. From Substep III.1 to Substep III.4, the map

$$\begin{cases} \widetilde{\Phi}_{\mathcal{V}} : [0,1]\times\widetilde{\mathbf{E}}_{\mathcal{V}} \to \mathbb{R}^\ell \\ (t,(v,g)) \mapsto \begin{cases} \widetilde{\Phi}^1_{\mathcal{V}}(4t,(v,g)) & \text{if} \quad t\in[0,\frac{1}{4}] \\ \widetilde{\Phi}^2_{\mathcal{V}}(4t-1,(v,g)) & \text{if} \quad t\in[\frac{1}{4},\frac{1}{2}] \\ \widetilde{\Phi}^3_{\mathcal{V}}(4t-2,(v,g)) & \text{if} \quad t\in[\frac{1}{2},\frac{3}{4}] \\ \widetilde{\Phi}^4_{\mathcal{V}}(4t-3,(v,g)) & \text{if} \quad t\in[\frac{3}{4},1]\end{cases}\end{cases}$$

defines a homotopy between $\widetilde{\Delta}_\partial$ and $\widetilde{\Delta}'$ such that for every $t\in[0,1]$, $(\widetilde{\Phi}_{\mathcal{V}}(t,\cdot,\cdot))^{-1}(0_{\mathbb{R}^\ell}) = \mathcal{V}^0$.

Step IV. Construction of a homotopy $\widetilde{H}_{\mathcal{V}}$ between $\widetilde{D}$ and $\widetilde{D}'$ (2) — The dynamics equivalence theorem. Following the remark made in Step II, since $\widetilde{\Phi}_{\mathcal{V}}$ is a homotopy between $\widetilde{\Delta}_\partial$ and $\widetilde{\Delta}'$ such that for every $t\in[0,1]$, $(\widetilde{\Phi}_{\mathcal{V}}(t,\cdot,\cdot))^{-1}(0_{\mathbb{R}^\ell}) = \mathcal{V}^0$ (cf. Step III), then the map

$$\begin{cases} \widetilde{H}^2_{\mathcal{V}} : [0,1]\times\widetilde{\mathbf{E}}_{\mathcal{V}} \to \mathbb{R}^\ell \\ (t,(v,g)) \mapsto (\widetilde{\Phi}_{\mathcal{V}}(t,\cdot,\cdot)\circ\theta_{\mathcal{V}}(1,\cdot,\cdot))(v,g)\end{cases}$$

defines a homotopy between $\widetilde{D}_\partial$ and $\widetilde{D}'$ such that for every $t\in[0,1]$, the map $\widetilde{H}^2_{\mathcal{V}}(t,\cdot,\cdot)$ is an extended network dynamics on $\mathcal{V}$. Finally, from Step I, the map

$$\begin{cases} \widetilde{H}_{\mathcal{V}} : [0,1]\times\widetilde{\mathbf{E}}_{\mathcal{V}} \to \mathbb{R}^\ell \\ (t,(v,g)) \mapsto \begin{cases} \widetilde{H}^1_{\mathcal{V}}(2t,(v,g)) & \text{if} \quad t\in[0,\frac{1}{2}] \\ \widetilde{H}^2_{\mathcal{V}}(2t-1,(v,g)) & \text{if} \quad t\in[\frac{1}{2},1]\end{cases}\end{cases}$$

is the required homotopy between $\widetilde{D}$ and $\widetilde{D}'$, such that for every $t\in[0,1]$, the map $\widetilde{H}_{\mathcal{V}}(t,\cdot,\cdot)$ is an extended network dynamics on $\mathcal{V}$. This step concludes the proof of the dynamics equivalence theorem. □

Appendix E. Proof of the indices equality theorem

We provide in Table E.4 a list of the notations used throughout the proof.

Table E.4
Table of notations for the indices equality theorem.

$\widetilde{\mathbf{E}}_{\mathcal{V}}$	*ε-ambient space $\mathcal{V}\times\widetilde{\mathbf{G}}$ associated with a set $\mathcal{V}$ of societies*
$\mathrm{T}\widetilde{\mathbf{G}}$	*Tangent bundle of $\widetilde{\mathbf{G}}$*
$\widetilde{H}_{\mathcal{V}} : [0,1]\times\widetilde{\mathbf{E}}_{\mathcal{V}} \to \mathbb{R}^{\ell}$	*Homotopy between two strongly inward-pointing extended network dynamics $\widetilde{D}$ and $\widetilde{D}'$ of the dynamics equivalence theorem, where $\mathcal{V}$ is a regular set of societies*
$\mathfrak{Z}(\widetilde{D})$	*Set $\{(v,g)\in\widetilde{\mathbf{E}}_{\mathcal{V}} : g\in\mathfrak{Z}(\widetilde{D}_v)\}$, with $\mathfrak{Z}(\widetilde{D}_v)$ being the set of zeros of the vector field $\widetilde{D}_v$, where $\widetilde{D}$ is an extended network dynamics*
$\zeta_{\mathfrak{c}} : \mathrm{T}\widetilde{\mathbf{G}} \to \widetilde{\mathbf{G}}\times\mathbb{R}^{\ell}$	*Canonical vector bundle isomorphism from Remark A.1*
$\mathfrak{X}(\widetilde{\mathbf{G}})$	*Set of vector fields on $\widetilde{\mathbf{G}}$*
$\Psi_{\mathfrak{c}} : C^0(\widetilde{\mathbf{G}},\mathbb{R}^{\ell}) \to \mathfrak{X}(\widetilde{\mathbf{G}})$	*Canonical bijection from Corollary 6.1 considering the vector bundle isomorphism $\zeta_{\mathfrak{c}}$*

Consider a fixed regular set $\mathcal{V}$ of societies (recall that $\varepsilon > 0$ and $g^0 \in \mathbf{G}$ are fixed). Additionally, consider two strongly inward-pointing extended network dynamics $\widetilde{D}, \widetilde{D}' : \widetilde{\mathbf{E}}_{\mathcal{V}} \to \mathrm{T}\widetilde{\mathbf{G}}$ on $\mathcal{V}$ (note that we do not use here the identification described in Section 6), and let $v \in \mathcal{V}$ be a society.

Recall that $\widetilde{H}_{\mathcal{V}} : \widetilde{\mathbf{E}}_{\mathcal{V}} : [0,1]\times\widetilde{\mathbf{E}}_{\mathcal{V}} \to \mathrm{T}\widetilde{\mathbf{G}}$ is the homotopy between $\widetilde{D}$ and $\widetilde{D}'$ from the dynamics equivalence theorem (see Section 6.1). Throughout this proof, we will simply denote this homotopy by $\widetilde{H}$.

Step I. Construction of a homotopy $\widetilde{H}_v$ between $\widetilde{D}_v$ and $\widetilde{D}'_v$. Observe that the map

$$\widetilde{H}_v : (t,\gamma)\in[0,1]\times\widetilde{\mathbf{G}} \mapsto \widetilde{H}(t,(v,\gamma)) \in \mathrm{T}\widetilde{\mathbf{G}}$$

defines a homotopy between $\widetilde{D}_v$ and $\widetilde{D}'_v$ such that for every $t\in[0,1]$,

$$\mathfrak{Z}(\widetilde{H}_v(t,\cdot)) = \mathfrak{Z}(\widetilde{D}_v) = \mathfrak{Z}(\widetilde{D}'_v) = \mathbf{P}_v, \tag{E.1}$$

where $\mathbf{P}_v = \{\gamma \in \mathbf{G} : \gamma \text{ is pairwise stable with respect to } v\}$. This follows from the definition of $\widetilde{H}$, since for every $t\in[0,1]$, $\widetilde{H}(t,\cdot,\cdot)$ is an extended network dynamics on $\mathcal{V}$.

Step II. Computation of the indices of $\widetilde{D}_v$ and $\widetilde{D}'_v$ at an isolated zero. Let $g\in\widetilde{\mathbf{G}}$ be an isolated zero of $\widetilde{D}_v$, and let (U,ϕ) be an interior chart of $\widetilde{\mathbf{G}}$ such that $g\in U$. Since $g\in\mathbf{G}$ and $\mathbf{G}\subset\mathrm{int}(\widetilde{\mathbf{G}})$, it follows that g is necessarily an interior zero of $\widetilde{D}_v$. From Eq. (E.1), we observe that g is also an interior isolated zero of $\widetilde{D}'_v$, and of $\widetilde{H}_v(t,\cdot)$ for each $t\in[0,1]$. The index of $\widetilde{D}_v$ at g corresponds to the index of its pushforward

$$\phi_*\widetilde{D}_v : x\in\phi(U) \mapsto (\mathrm{T}_{\phi^{-1}(x)}\phi\circ\widetilde{D}_v\circ\phi^{-1})(x) \in \mathrm{T}\phi(U)$$

by ϕ at $\phi(g)$, which in turn corresponds to the index of the map

$$\Psi_{\mathfrak{c}}^{-1}(\phi_*\widetilde{D}_v) : \phi(U) \to \mathbb{R}^{\ell}$$

at $\phi(g)$. This map is simply denoted by $\Psi_*\widetilde{D}_v$. Finally, its index is given by the topological degree of the function

$$\begin{cases} (\Psi_*\widetilde{D}_v)^{\phi(g)} : & \partial\overline{B}(\phi(g),\epsilon) \to \mathbb{S}^{\ell-1} \\ & x \mapsto \frac{\Psi_*\widetilde{D}_v(x)}{\|\Psi_*\widetilde{D}_v(x)\|_2} \end{cases},$$

where $\epsilon>0$ is chosen such that

$$\overline{B}(\phi(g),\epsilon)\cap\left((\Psi_*\widetilde{D}_v)^{-1}(0_{\mathbb{R}^{\ell}})\backslash\{\phi(g)\}\right) = \emptyset. \tag{E.2}$$

Since g is an isolated zero, such a value of ϵ exists and is fixed from now on (see Appendix A.3 for some reminders on differential geometry).

From Eq. (E.1), the same ϵ also satisfies

$$\overline{B}(\phi(g),\epsilon)\cap\left((\Psi_*\widetilde{D}'_v)^{-1}(0_{\mathbb{R}^{\ell}})\backslash\{\phi(g)\}\right) = \emptyset,$$

and even

$$\overline{B}(\phi(g),\epsilon)\cap\left((\Psi_*\widetilde{H}_v(t,\cdot))^{-1}(0_{\mathbb{R}^{\ell}})\backslash\{\phi(g)\}\right) = \emptyset,$$

for every $t\in[0,1]$, where $\Psi_*\widetilde{D}'_v$ (resp. $\Psi_*\widetilde{H}_v(t,\cdot)$) denotes $\Psi_{\mathfrak{c}}^{-1}(\phi_*\widetilde{D}'_v)$ (resp. $\Psi_{\mathfrak{c}}^{-1}(\phi_*\widetilde{H}_v(t,\cdot))$). Indeed, for every $t\in[0,1]$, we have

$$\mathfrak{Z}(\phi_*\widetilde{H}_v(t,\cdot)) = \mathfrak{Z}(\phi_*\widetilde{D}_v) = \mathfrak{Z}(\phi_*\widetilde{D}'_v) = \phi(\mathbf{P}_v\cap U), \tag{E.3}$$

since for every $x\in\phi(U)$, the tangent map $\mathrm{T}_{\phi^{-1}(x)}\phi$ of ϕ at $\phi^{-1}(x)$ being a vector space isomorphism (as $\phi : U\to\phi(U)$ is, by definition, a diffeomorphism). Consequently, its kernel consists only of the zero vector $\overrightarrow{0_{\phi^{-1}(x)}}$ in $\mathrm{T}_{\phi^{-1}(x)}\widetilde{\mathbf{G}}$. Finally, from Remark A.1, for every $t\in[0,1]$,

$$(\Psi_*\widetilde{H}_v(t,\cdot))^{-1}(0_{\mathbb{R}^{\ell}}) = \mathfrak{Z}(\phi_*\widetilde{H}_v(t,\cdot)). \tag{E.4}$$

Using Eq. (E.2), Eq. (E.3) and Eq. (E.4) we obtain the desired result: in particular, for $t=1$, the index of $\widetilde{D}'_v$ at g is equal to the topological degree of the map

$$\begin{cases} (\Psi_*\widetilde{D}'_v)^{\phi(g)} : & \partial\overline{B}(\phi(g),\epsilon) \to \mathbb{S}^{\ell-1} \\ & x \mapsto \frac{\Psi_*\widetilde{D}'_v(x)}{\|\Psi_*\widetilde{D}'_v(x)\|_2} \end{cases}.$$

Step III. Construction of a homotopy $\widetilde{H}_v^{\phi(g)}$ between $(\Psi_*\widetilde{D}_v)^{\phi(g)}$ and $(\Psi_*\widetilde{D}'_v)^{\phi(g)}$. Let us define the maps

$$\widetilde{H}_v^1 : (t,x)\in[0,1]\times\phi(U) \mapsto \phi_*\widetilde{H}_v(t,x) = (\mathrm{T}_{\phi^{-1}(x)}\phi\circ\widetilde{H}_v(t,\cdot)\circ\phi^{-1})(x) \in \mathrm{T}\phi(U)$$

and

$$\widetilde{H}_v^2 : (t,x)\in[0,1]\times\phi(U) \mapsto (\mathrm{pr}_{\mathbb{R}^{\ell}}\circ\zeta_{\mathfrak{c}}\circ\widetilde{H}_v^1(t,\cdot))(x) \in \mathbb{R}^{\ell},$$

where $\zeta_{\mathfrak{c}}$ is defined in Remark A.1 (see Appendix A.3). Note that $\widetilde{H}_v^1$ is a homotopy between the pushforward $\phi_*\widetilde{D}_v$ of $\widetilde{D}_v$ by ϕ and the pushforward $\phi_*\widetilde{D}'_v$ of $\widetilde{D}'_v$ by ϕ. It is continuous (since it is the composition of two continuous maps), and for every $x\in\phi(U)$, $\widetilde{H}_v^1(0,x) = \phi_*\widetilde{D}_v(x)$ and $\widetilde{H}_v^1(1,x) = \phi_*\widetilde{D}'_v(x)$. Similarly, observe that $\widetilde{H}_v^2$ is a homotopy between $\Psi_*\widetilde{D}_v = \mathrm{pr}_{\mathbb{R}^{\ell}}\circ\zeta_{\mathfrak{c}}\circ\phi_*\widetilde{D}'_v$ and $\Psi_*\widetilde{D}_v = \mathrm{pr}_{\mathbb{R}^{\ell}}\circ\zeta_{\mathfrak{c}}\circ\phi_*\widetilde{D}'_v$.

Now, notice that the map

$$\begin{cases} \widetilde{H}_v^{\phi(g)} : & [0,1]\times\partial\overline{B}(\phi(g),\epsilon) \to \mathbb{S}^{\ell-1} \\ & (t,x) \mapsto \frac{\widetilde{H}_v^2(t,x)}{\|\widetilde{H}_v^2(t,x)\|_2} \end{cases}$$

is a well-defined homotopy between $(\Psi_*\widetilde{D}_v)^{\phi(g)}$ and $(\Psi_*\widetilde{D}'_v)^{\phi(g)}$. To demonstrate this, let us prove that $\left([0,1]\times\overline{B}(\phi(g),\epsilon)\right)\cap\left((\widetilde{H}_v^2)^{-1}(0_{\mathbb{R}^{\ell}})\backslash([0,1]\times\{\phi(g)\})\right)$ is empty. By contradiction, suppose that there exists $(t,x)\in[0,1]\times\overline{B}(\phi(g),\epsilon)$ which also belongs to $(\widetilde{H}_v^2)^{-1}(0_{\mathbb{R}^{\ell}})\backslash([0,1]\times\{\phi(g)\})$. By definition,

$$\begin{aligned} &(\widetilde{H}_v^2)^{-1}(0_{\mathbb{R}^{\ell}}) \\ &= \{(t,x)\in[0,1]\times\phi(U) : (\mathrm{pr}_{\mathbb{R}^{\ell}}\circ\zeta_{\mathfrak{c}}\circ\widetilde{H}_v^1(t,\cdot))(x) = 0_{\mathbb{R}^{\ell}}\} \\ &= \{(t,x)\in[0,1]\times\phi(U) : (\mathrm{pr}_{\mathbb{R}^{\ell}}\circ\zeta_{\mathfrak{c}}\circ\mathrm{T}_{\phi^{-1}(x)}\phi\circ\widetilde{H}_v(t,\cdot)\circ\phi^{-1})(x) = 0_{\mathbb{R}^{\ell}}\}. \end{aligned}$$

Therefore, by definition of $\zeta_{\mathfrak{c}}$ (cf. Remark A.1) and the projection $\mathrm{pr}_{\mathbb{R}^{\ell}}$, (t,x) belongs to $(\widetilde{H}_v^2)^{-1}(0_{\mathbb{R}^{\ell}})\backslash([0,1]\times\{\phi(g)\})$ if and only if $x\neq\phi(g)$ and $(\mathrm{T}_{\phi^{-1}(x)}\phi\circ\widetilde{H}_v(t,\cdot)\circ\phi^{-1})(x) = \phi_*\widetilde{H}_v(t,x) = \overrightarrow{0_x}$, where $\overrightarrow{0_x}$ denotes the zero vector in $\mathrm{T}_x\phi(U)$. However, recalling from Eq. (E.2), Eq. (E.3) and Eq. (E.4), that for every $t'\in[0,1]$,

$$\mathfrak{Z}(\phi_*\widetilde{H}_v(t',\cdot)) = \mathfrak{Z}(\mathrm{T}_{\phi^{-1}(x)}\phi\circ\widetilde{H}_v(t',\cdot)\circ\phi^{-1}) = \phi(\mathbf{P}_v\cap U) = \mathfrak{Z}(\phi_*\widetilde{D}_v).$$

Furthermore, as explained above and by definition of ϵ,

$$\overline{B}(\phi(g),\epsilon)\cap(\mathfrak{Z}(\phi_*\widetilde{D}_v)\backslash\{\phi(g)\}) = \emptyset,$$

which implies that for every $t'\in[0,1]$,

$$\overline{B}(\phi(g),\epsilon)\cap(\mathfrak{Z}(\phi_*\widetilde{H}_v(t',\cdot))\backslash\{\phi(g)\}) = \emptyset.$$

In particular, $\overline{B}(\phi(g),\epsilon)\cap(\mathfrak{Z}(\phi_*\widetilde{H}_v(t,\cdot))\backslash\{\phi(g)\})$ is empty, which contradicts the fact that (t,x) belongs to the set $(\widetilde{H}_v^2)^{-1}(0_{\mathbb{R}^{\ell}})\backslash([0,1]\times\{\phi(g)\})$.

Step IV. The indices equality theorem. Now, from Proposition A.1, observe that it is sufficient to show that the maps $(\Psi_*\widetilde{D}_v)^{\phi(g)}$ and $(\Psi_*\widetilde{D}'_v)^{\phi(g)}$ are homotopic in order to conclude that $\mathrm{ind}_g(\widetilde{D}_v) = \mathrm{ind}_g(\widetilde{D}'_v)$. Therefore, from Step III, because

$$\begin{cases} \widetilde{H}_v^{\phi(g)} : & [0,1]\times\partial\overline{B}(\phi(g),\epsilon) \to \mathbb{S}^{\ell-1} \\ & (t,x) \mapsto \frac{\widetilde{H}_v^2(t,x)}{\|\widetilde{H}_v^2(t,x)\|_2} \end{cases}$$

is a well-defined homotopy between $(\Psi_* \widetilde{D}_v)^{\phi(g)}$ and $(\Psi_* \widetilde{D}'_v)^{\phi(g)}$, this implies that $\mathrm{ind}_g(\widetilde{D}_v) = \mathrm{ind}_g(\widetilde{D}'_v)$. This step concludes the proof of the indices equality theorem. □

Appendix F. Proof of Proposition 6.2

Let r be the projection from $\widetilde{\mathbf{G}}$ to the closest point in $\mathbf{G}$ (this map is a retraction of $\widetilde{\mathbf{G}}$ on $\mathbf{G}$, i.e., a continuous map such that its composition with the inclusion $\mathbf{G} \hookrightarrow \widetilde{\mathbf{G}}$ is the identity map on $\mathbf{G}$), $b : \widetilde{\mathbf{G}} \to [0,1]$ be a continuous map such that $b(\mathbf{G}) = 1$ and $b(\partial\widetilde{\mathbf{G}}) = 0$, and $X : \mathbf{G}^\varepsilon = \widetilde{\mathbf{G}} \to \mathbb{R}^\ell$ be a (continuous) vector field on $\mathbf{G}^\varepsilon$ such that $X(\mathbf{G}) = 0_{\mathbb{R}^\ell}$ and for every $\varepsilon' \in (0,\varepsilon]$, every $g \in \partial\mathbf{G}^{\varepsilon'}$, and every $x \in \mathrm{N}_{\mathbf{G}^{\varepsilon'}}(g)$, $\langle X(g), x\rangle < 0$. Consider the map

$$\begin{cases} D^\varepsilon : & \mathbf{E}^\varepsilon_{\mathcal{V}} & \to & \mathbb{R}^\ell \\ & (v,g) & \mapsto & b(g)D(v,r(g)) + X(g) \end{cases}.$$

First, observe that for every $(v,g) \in \mathbf{E}_{\mathcal{V}}$,

$$D^\varepsilon(v,g) = b(g)D(v,r(g)) + X(g) = D(v,g).$$

Indeed, since $g \in \mathbf{G}$, we have $b(g) = 1$, $r(g) = g$ and $X(g) = 0$, which implies that

$$\mathbf{P}_{\mathcal{V}} \subset (D^\varepsilon)^{-1}(0_{\mathbb{R}^\ell}). \tag{F.1}$$

Second, observe that $D^\varepsilon = b \times (D \circ (\mathrm{id}_{\mathcal{V}} \times r)) + X$ is a continuous map, since D, b, r and X are continuous.

Last, for every $v \in \mathcal{V}$, every $\varepsilon' \in (0,\varepsilon]$, every $g \in \partial\mathbf{G}^{\varepsilon'}$ and every $x \in \mathrm{N}_{\mathbf{G}^{\varepsilon'}}(g)$,

$$\langle D^\varepsilon_v(g), x\rangle = \langle b(g)D(v,r(g)) + X(g), x\rangle = b(g)\langle D(v,r(g)), x\rangle + \langle X(g), x\rangle < 0.$$

Indeed: (i) by definition, $b(g) \in [0,1]$; (ii) $\langle D(v,r(g)), x\rangle \leq 0$ because $r(g) \in \partial\mathbf{G}$ (by definition) and because D is inward-pointing (by assumption); (iii) by definition, $\langle X(g), x\rangle < 0$. Therefore, for every $v \in \mathcal{V}$, D^ε_v has no zeros on $\mathbf{G}^\varepsilon \backslash \mathbf{G}$, which finally implies that

$$(D^\varepsilon)^{-1}(0_{\mathbb{R}^\ell}) = \mathbf{P}_{\mathcal{V}}$$

(using Eq. (F.1)). □

Data availability

No data was used for the research described in the article.

References

Bala, Venkatesh, Goyal, Sanjeev, 2000. A noncooperative model of network formation. Econometrica.

Bich, Philippe, Fixary, Julien, 2022. Network formation and pairwise stability: A new oddness theorem. J. Math. Econom..

Bich, Philippe, Fixary, Julien, 2024. Oddness of the number of Nash equilibria: The case of polynomial payoff functions. Games Econom. Behav..

Bich, Philippe, Morhaim, Lisa, 2020. On the existence of pairwise stable weighted networks. Math. Oper. Res..

Blume, Lawrence E., Zame, William R., 1994. The algebraic geometry of perfect and sequential equilibrium. Econometrica.

Bochnak, Jacek, Coste, Michel, Roy, Marie-Françoise, 1998. Real Algebraic Geometry. Springer.

Bott, Raoul, Tu, Loring W., 1982. Differential Forms in Algebraic Topology. Springer.

Bramoullé, Yann, Kranton, Rachel, 2007. Public goods in networks. J. Econom. Theory.

Calvó-Armengol, Antoni, İlkılıç, Rahmi, 2009. Pairwise stability and Nash equilibria in network formation. Int. J. Game Theory.

Demichelis, Stefano, Germano, Fabrizio, 2000a. On the indices of zeros of Nash fields. J. Econom. Theory.

Demichelis, Stefano, Germano, Fabrizio, 2000b. Some consequences of the unknottedness of the walras correspondence. J. Math. Econom..

Demichelis, Stefano, Germano, Fabrizio, 2002. On (un)knots and dynamics in games. Games Econom. Behav..

Dold, Albrecht, 1995. Lectures on Algebraic Topology. Springer.

Govindan, Srihari, McLennan, Andrew, 2001. On the generic finiteness of equilibrium outcome distributions in game forms. Econometrica.

Govindan, Srihari, Wilson, Robert, 1997. Equivalence and invariance of the index and degree of Nash equilibria. Games Econom. Behav..

Govindan, Srihari, Wilson, Robert, 2001. Direct proofs of generic finiteness of Nash equilibrium outcomes. Econometrica.

Guillemin, Victor, Pollack, Alan, 1975. Differential Topology. Pearson.

Gul, Faruk, Pearce, David, Stacchetti, Ennio, 1993. A bound of the proportion of pure strategy equilibria in generic games. Math. Oper. Res..

Harsanyi, John, 1973. Oddness of the number of equilibrium points: A new proof. Int. J. Game Theory.

Herings, P. Jean-Jacques, Peeters, Ronald J.A.P., 2001. A differentiable homotopy to compute Nash equilibria of n-person games. Econom. Theory.

Jackson, Matthew O., Watts, Alison, 2001. The existence of pairwise stable networks. Seoul J. Econ..

Jackson, Matthew O., Watts, Alison, 2002. The evolution of social and economic networks. J. Econom. Theory.

Jackson, Matthew O., Wolinsky, Asher, 1996. A strategic model of social and economic networks. J. Econom. Theory.

Kohlberg, Elon, Mertens, Jean-François, 1986. On the strategic stability of equilibria. Econometrica.

Laraki, Rida, Renault, Jérôme, Sorin, Sylvain, 2019. Mathematical Foundations of Game Theory. Springer.

Lee, John M., 2012. Introduction To Smooth Manifolds. Springer.

Mas-Colell, Andreu, 2010. Generic finiteness of equilibrium payoffs for bimatrix games. J. Math. Econom..

Predtetchinski, Arkadi, 2009. A general structure theorem for the Nash equilibrium correspondence. Games Econom. Behav..

Ritzberger, Klaus, 1994. The theory of normal form games from the differentiable viewpoint. Int. J. Game Theory.

Rockafellar, Tyrrell R., Wets, Roger J.B., 1998. Variational Analysis. Springer.

Watts, Alison, 2001. A dynamic model of network formation. Games Econom. Behav..

Wilson, Robert, 1971. Computing equilibria of n-person games. SIAM J. Appl. Math..